

# Platformless AI: A Decentralised Protocol for Trustless AI Inference

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## Abstract

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Platformless AI is a decentralised protocol that enables trustless AI inference, multi-agent orchestration, and agentic AI workflows through a peer-to-peer marketplace of independent compute providers. By combining Ethereum smart contracts, cryptographic proofs, and decentralised storage, the protocol eliminates platform intermediaries while maintaining accountability, privacy, and fair compensation. Beyond chat-based inference, the protocol supports autonomous AI agents — coding assistants, multi-tool workflows, and SaaS AI backends — through both an Anthropic Messages API compatibility layer (Claude Bridge) and an OpenAI Chat Completions API compatibility layer (OpenAI Bridge), enabling any Anthropic-compatible or OpenAI-compatible client to run on decentralised infrastructure. The protocol also supports **fine-tuning as a marketplace job**: a user's dataset trains a LoRA adapter on a rented GPU with per-slice on-chain settlement, so a run that stops halfway bills only for the work done. The dataset is encrypted in transit and at rest and the adapter is returned encrypted, but the host decrypts both in order to train; extending the confidential-computing path described below to training is what would make it host-blind, and a fine-tune has already been run on that confidential-computing host, with the dataset decrypted only inside the confidential VM and two independent runs producing byte-identical adapters. It supports image generation via host-side diffusion model sidecars, AI video generation across thirteen modes — text-to-video, image-to-video, first-last-frame, reference-conditioned generation, video restyling (IC-LoRA union control, where the motion and camera of a control clip drive a newly styled output), outpainting to a new aspect ratio, prompt-driven video editing, restore, 2x upscale, water, day-to-night, novel-view synthesis, and SDR-to-HDR reconstruction — via ComfyUI sidecars running LTX 2.3 (with pinned-template provenance verified client-side and anchored on-chain at settlement, so buyers can verify exactly which inputs, including any control video, produced the clip, and hold the host to a signed, on-chain commitment to the model and template it ran), all of them driven from a native Blender extension that puts paid, provenance-bound generation inside a professional editing timeline, GPU-accelerated video transcoding with end-to-end encryption, HLS adaptive bitrate streaming with per-segment encryption, load balancing across multiple hosts, and experimental multi-agent orchestration via Google's Agent-to-Agent (A2A) protocol (v1.0.0-rc) — decomposing complex goals into task graphs, routing sub-tasks to optimal models across decentralised GPU hosts, and settling payments on-chain with cryptographic proof collection. Platformless AI is, to our knowledge, the first DePIN project to implement the A2A protocol. For privacy-critical workloads, a confidential-computing path, wired into the node's live request path and proven end to end on real confidential-computing hardware (Intel TDX with an NVIDIA H200, September 2026), extends the protocol into Trusted Execution Environments, where the model and the user's inputs are decrypted only inside a confidential VM whose memory the host operator cannot read. This whitepaper describes the technical architecture, economic model, and security mechanisms that enable a truly

open AI marketplace where anyone can provide or consume AI services without centralised gatekeepers.

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## 1. Introduction

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The AI revolution has brought unprecedented capabilities to humanity, yet access to these capabilities remains concentrated in the hands of a few large technology companies. These platforms act as gatekeepers, controlling which models users can access, what they can be used for, and at what price. Users have no choice but to trust these intermediaries with their data, their queries, and their dependency on AI-powered workflows.

**Platformless AI** represents a paradigm shift: a protocol, not a platform. Like BitTorrent democratised file sharing by eliminating central servers, Platformless AI democratises AI inference by creating a trustless marketplace where independent compute providers—called **hosts**—offer AI services directly to users without intermediaries.

The protocol leverages:

- **Ethereum smart contracts** for transparent, automated settlement
- **End-to-end encryption** for privacy-preserving inference
- **Cryptographic commitments** (STARK-anchored) binding each job to the model, input, and output the host claims
- **Decentralised storage (S5/Sia)** for immutable audit trails
- **Economic incentives** aligned to ensure honest behaviour

The result is an open, permissionless AI marketplace where hosts compete on price and quality, users retain full control of their data, and trust is established through verifiable code, mathematics, and economic incentives rather than reputation.

Critically, this infrastructure extends beyond simple chat to support **autonomous AI agents, image generation, AI video generation** (thirteen modes: text-to-video, image-to-video, first-last-frame,

ingredients, restyle, outpaint, edit, restore, upscale, water, day-to-night, cross-view, and SDR-to-HDR — all with settled on-chain economics, and all drivable from inside Blender), **GPU-accelerated video transcoding**, and **multi-agent orchestration** (experimental). The Claude Bridge translates the Anthropic Messages API, and the OpenAI Bridge translates the OpenAI Chat Completions API, into the protocol's encrypted WebSocket layer, enabling tools like Claude Code, OpenCode, Continue, and Cursor to run entirely on decentralised GPU hosts. The transcoding pipeline enables end-to-end encrypted video processing across multiple independent hosts with capacity-aware load balancing, including HLS adaptive bitrate streaming where the transcoder produces fMP4 segments with per-segment encryption (preview segments unencrypted for free access, paid segments encrypted with XChaCha20-Poly1305) and the SDK generates M3U8 playlists client-side for playback via hls.js. Since NVIDIA's NVENC encoder is separate hardware from the CUDA cores used for inference, hosts can transcode video and run LLM inference simultaneously without performance degradation, enabling dual revenue streams from the same GPU. The orchestrator package ([@fabstir/orchestrator](#)) adds DAG-based task decomposition, intelligent model routing, and inter-agent collaboration via Google's Agent-to-Agent (A2A) protocol — an emerging standard (~11 months old, v1.0.0-rc) backed by 150+ technology partners and the Linux Foundation. The x402 HTTP payment protocol enables USDC micropayments for inter-agent delegation, running alongside the existing on-chain escrow model. This positions Platformless AI as infrastructure for the next generation of AI-powered applications, including agentic coding assistants, multi-agent workflows, video platforms, creative AI tools, and SaaS AI backends.

## 2. The Problem: Centralisation in AI

### 2.1 Platform Lock-in

Today's AI landscape is dominated by vertically integrated platforms that control the entire stack:

| Layer          | Centralised Control                         |
|----------------|---------------------------------------------|
| Models         | Platforms decide which models are available |
| Infrastructure | Users must use platform hardware            |
| Pricing        | Platforms set prices unilaterally           |
| Data           | All queries flow through platform servers   |
| Policies       | Platforms decide acceptable use             |

This creates **vendor lock-in** where users become dependent on a single provider's ecosystem, pricing, and policies.

### 2.2 Privacy Concerns

When users send prompts to centralised AI services:

- Prompts are transmitted and stored on platform servers
- Conversation history is retained indefinitely
- Data may be used for model training without explicit consent

- Sensitive queries expose business intelligence, personal information, or trade secrets

Users must trust platforms to handle this data responsibly—a trust that has been violated repeatedly across the technology industry.

## 2.3 Censorship and Availability

Centralised platforms can:

- Restrict access to certain models or capabilities
- Impose content policies that prevent legitimate use cases
- Discontinue services without notice
- Deny service based on geography, payment method, or arbitrary criteria

There is no recourse when a platform decides your use case is not welcome.

## 2.4 Economic Inefficiency

Platform economics favour the platform, not participants:

- Compute providers (if any external ones exist) receive a fraction of revenue
- Users pay premium prices that include platform overhead and profit margins
- No competition on pricing—platforms set prices
- Innovation is constrained to what platforms choose to offer

## 2.5 The Trust Problem

The fundamental issue is **forced trust**. Users must trust that:

- Platforms will not misuse their data
- Pricing is fair
- Services will remain available
- Quality will be maintained
- Terms will not change adversely

This trust cannot be verified, only assumed.

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## 3. The Solution: Platformless AI

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Platformless AI replaces platform-mediated trust with **cryptographic trust**. Instead of trusting a company, users trust verifiable code, cryptographic mathematics, and carefully aligned economic incentives.

### 3.1 Core Principles

#### Permissionless Participation

Anyone can become a host by staking tokens and registering their compute resources. No approval process, no gatekeepers. The only requirement is economic commitment (stake) that aligns host

incentives with honest behaviour.

### **Direct Peer-to-Peer Connections**

Users connect directly to hosts via WebSocket. There is no platform server routing traffic. The SDK can establish direct connections from desktop applications, mobile apps, or server environments. Browser applications use a minimal proxy only to bypass CORS restrictions—the proxy does not see decrypted content.

### **Blockchain-Based Coordination**

Smart contracts on Ethereum (specifically Base, an L2) handle:

- Host registration and discovery
- Session creation and payment escrow
- Proof verification and settlement
- Model governance and approval

The blockchain provides a single source of truth that all participants can verify.

### **End-to-End Encryption**

All inference traffic is encrypted using modern cryptographic primitives:

- **ECDH key exchange** for session key establishment
- **XChaCha20-Poly1305** for symmetric encryption
- **ECDSA signatures** for authentication
- **Forward secrecy** via ephemeral session keys

Even if a host's long-term key is compromised, past sessions cannot be decrypted.

### **Decentralised Storage**

Conversation history, proofs, and audit data are stored on **S5** (backed by the Sia network), a decentralised storage layer that provides:

- Content-addressed immutability
- Censorship resistance
- Distributed redundancy
- Cost-effective archival

Data stored on S5 is encrypted by the SDK before upload—S5 nodes cannot read content.

## **3.2 How It Works: A User's Perspective**

1. User authenticates with their Ethereum wallet
2. SDK discovers available hosts from the NodeRegistry contract
3. User selects a host based on price, model support, and reputation
4. User creates a session by depositing funds into the JobMarketplace contract
5. SDK establishes encrypted WebSocket connection to host
6. User sends prompts; host streams responses (all encrypted)
7. Host submits signed proofs of work to the contract
8. Session completes; contract settles payments automatically
9. Conversation is stored encrypted on S5 for user's records

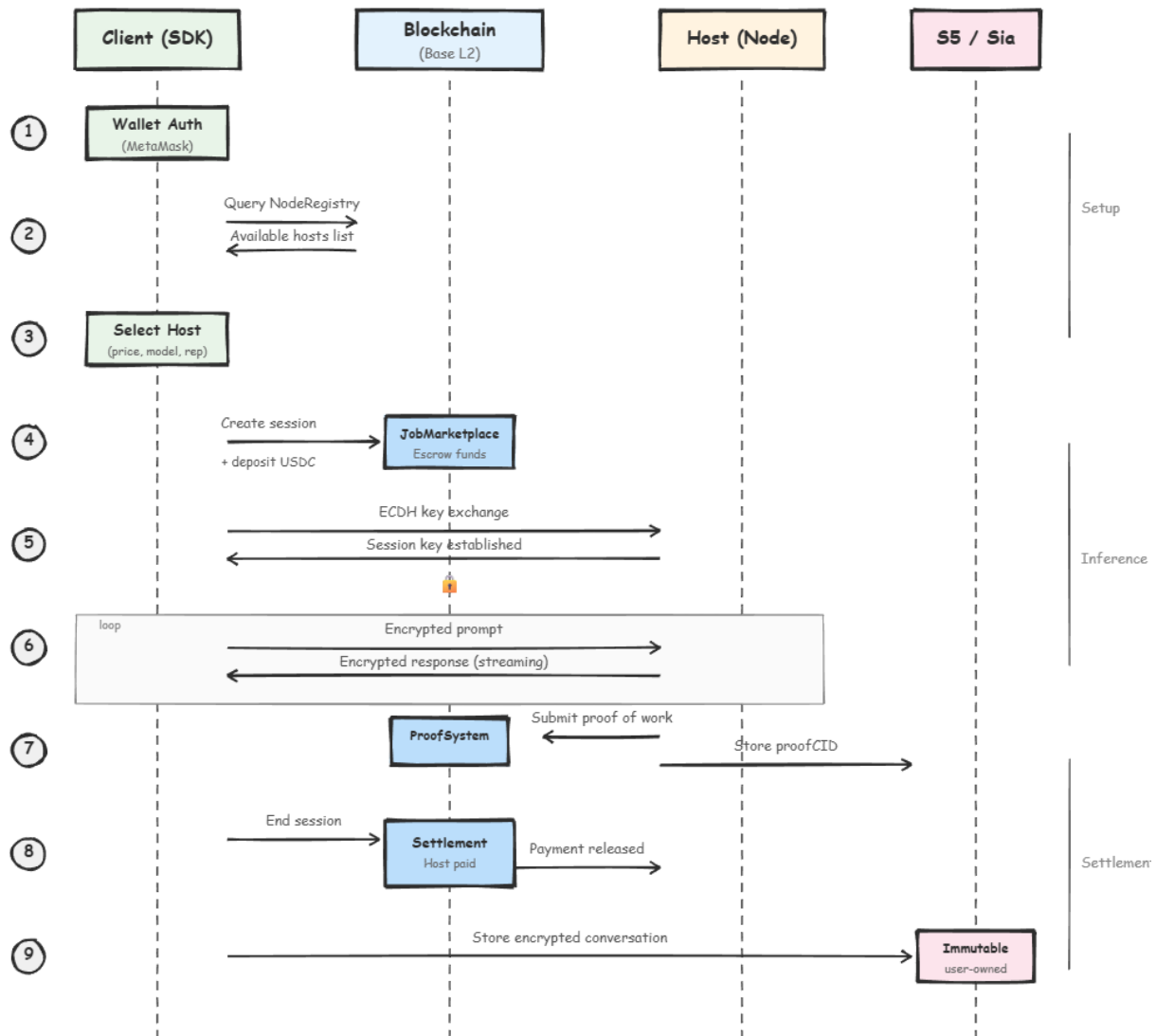


Figure 1: Session Lifecycle Flow showing the 9-step process from wallet authentication through to encrypted conversation storage.

The user never trusts a platform. Trust is established through:

- **Contracts** that enforce rules automatically
- **Encryption** that protects content
- **Proofs** that verify computation
- **Stakes** that deter malicious behaviour

### 3.3 How It Works: A Host's Perspective

1. Host stakes FAB tokens in the NodeRegistry contract
2. Host registers metadata (API URL, supported models, pricing)
3. Host runs inference server (GPU hardware + Fabstir Node software)
4. When sessions are created, host's server accepts WebSocket connections
5. Host processes encrypted prompts, streams encrypted responses
6. Host submits signed proofs of work periodically
7. On session completion, contract releases payment to HostEarnings
8. Host withdraws accumulated earnings at their convenience

Hosts operate as independent businesses. They set their own prices, choose which models to support, and manage their own infrastructure. The protocol handles discovery, payment, and dispute resolution.

## 4. Technical Architecture

### 4.1 System Overview

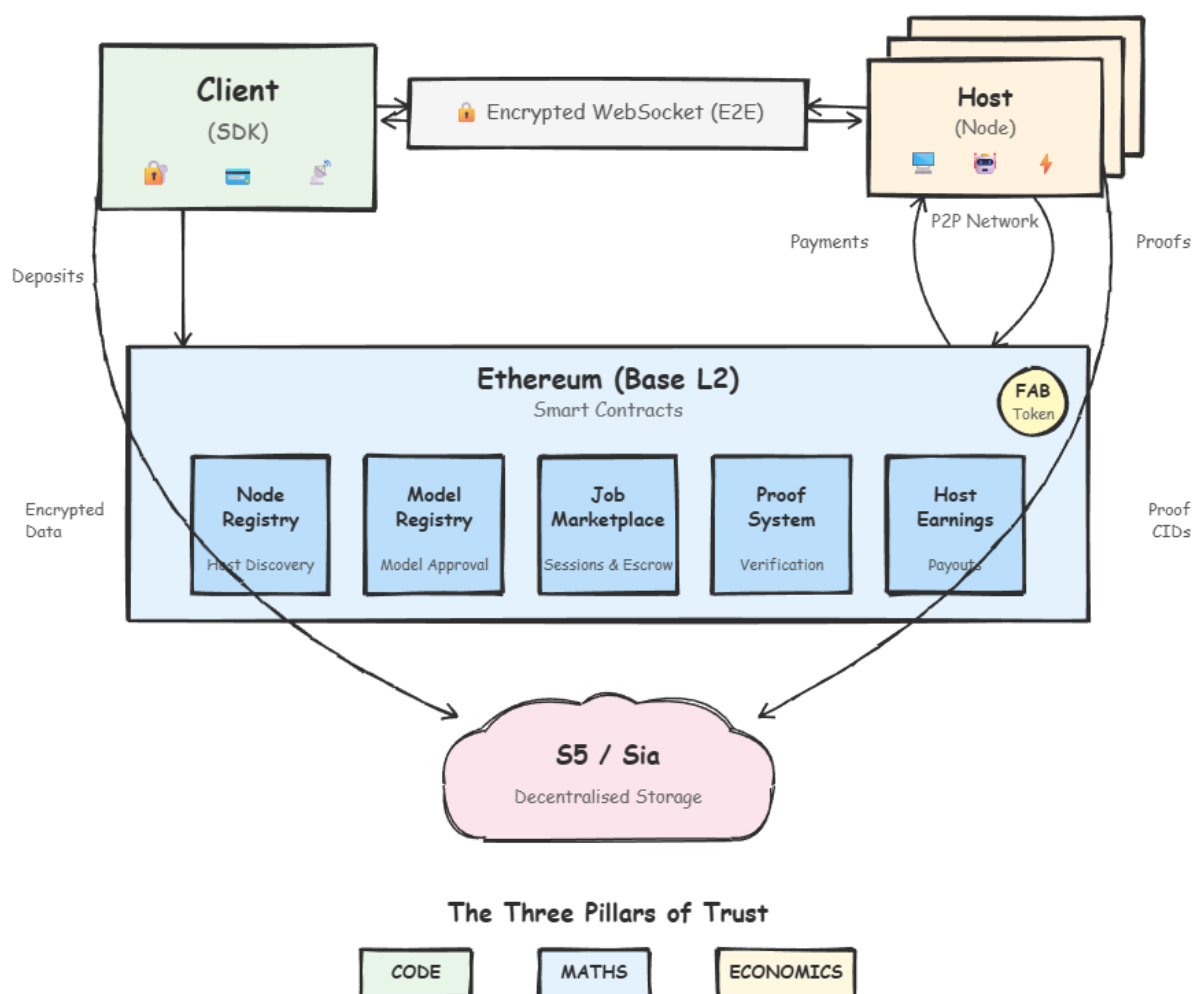


Figure 2: High-Level System Architecture showing client-host P2P connections, blockchain coordination layer (Base L2), and decentralised S5 storage.

The architecture comprises four main layers:

- **Client Layer:** SDK-powered applications connecting via encrypted WebSocket
- **Host Layer:** P2P network of independent GPU providers running inference
- **Blockchain Layer:** Ethereum (Base L2) smart contracts for coordination, escrow, and settlement
- **Storage Layer:** S5/Sia decentralised storage for encrypted conversations and proofs

## 4.2 Smart Contracts

The protocol uses **UUPS upgradeable proxy contracts** for future improvements without disrupting existing sessions.

### NodeRegistry

Manages host registration, staking, and pricing.

```
struct Node {
    address operator;          // Host's Ethereum address
    uint256 stakedAmount;      // FAB tokens staked (min 1000)
    bool active;               // Currently accepting sessions
    string metadata;           // JSON (name, description, etc.)
    string apiUrl;             // Inference endpoint URL
    bytes32[] models;          // Supported model IDs
    uint256 minPriceNative;     // Minimum price for ETH payments
    uint256 minPriceStable;    // Minimum price for USDC payments
}
```

Key functions:

- **registerNode()** - Stake tokens and register as host
- **unregisterNode()** - Withdraw stake and deregister
- **setModelPricing()** - Set per-model pricing
- **getNodesForModel()** - Discover hosts for a model

### JobMarketplace

Handles session lifecycle and payment escrow.

```
struct SessionJob {
    address depositor;         // User who deposited funds
    address host;              // Host providing inference
    address paymentToken;      // ETH (address(0)) or USDC
    uint256 deposit;           // Escrowed funds
    uint256 pricePerToken;     // Agreed price
    uint256 tokensUsed;        // Tokens consumed
    SessionStatus status;      // Active, Completed, TimedOut
    string conversationCID;    // S5 storage reference
}
```

Key functions:

- `createSessionJob()` - Create session and escrow funds
- `submitProofOfWork()` - Host submits signed proof
- `completeSessionJob()` - Settle and distribute payments

*Note: The contract includes a **DISPUTE\_WINDOW** parameter (configurable via upgrade) that delays host-initiated session completion. This window allows users to review session outcomes before final settlement. The optimal duration is being evaluated—currently set to 30 seconds for testing, with production values to be determined based on user experience research.*

## ModelRegistry

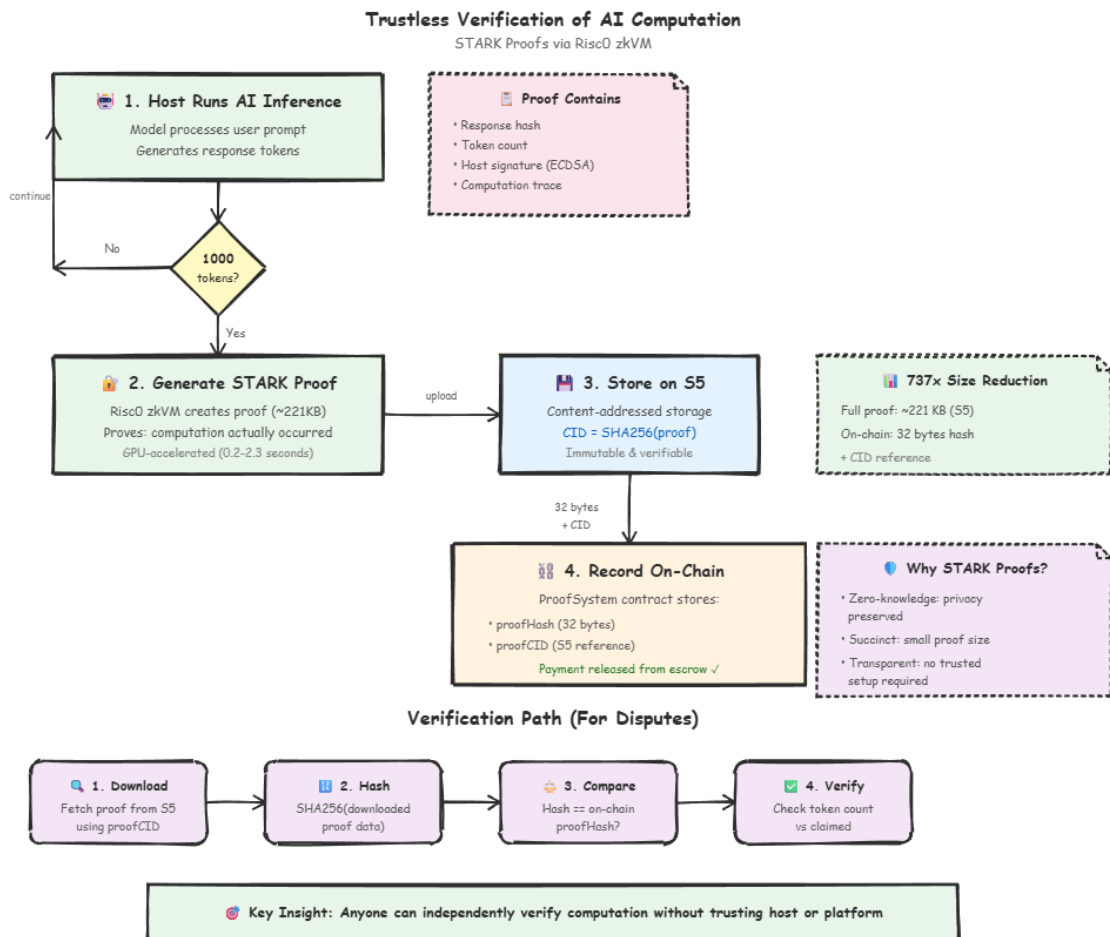
Governs which models can be used in the marketplace.

- Community-governed model approval
- Model ID derivation from repository/filename
- Prevents malicious model distribution

## ProofSystem

Verifies host signatures and integrates with zero-knowledge proofs.

- ECDSA signature verification
- STARK proof hook (future: on-chain verification)
- Graceful degradation when proof system unavailable



*Figure 3: Proof-of-Inference Flow, from inference through STARK proof generation, S5 storage, and on-chain recording, with the verification path for disputes.*

### What the proof establishes, and what it does not

The protocol is precise about this, because the distinction matters and the industry is often loose with it.

The zkVM guest reads four 32-byte hashes — `jobId`, `modelHash`, `inputHash`, `outputHash` — and commits them to its journal. The resulting STARK therefore establishes a **tamper-evident, on-chain-anchored commitment** to *which model the host claims it ran, over which input, to produce which output*. It does **not** re-execute the inference inside the zkVM, and it does not prove that the committed output is what that model would produce from that input. Proving transformer inference at 20-billion-parameter scale inside a zkVM is not something the state of the art can do today, at any price, and the protocol does not pretend otherwise.

What that commitment is worth in practice:

- **The client checks it against data it already holds.** It knows the prompt it sent and the output it received, so it recomputes `inputHash` and `outputHash` itself. A host that substitutes either is caught immediately, by arithmetic, with no trust required.
- **The commitment is immutable and attributable.** It is signed, anchored on-chain, and staked against. A host that commits to one model and serves another has produced durable, signed evidence of the discrepancy, which is what makes evidence-based slashing (Section 9) possible at all.
- **What it cannot do is bind the weights.** `modelHash` is what the host *says* it loaded. Closing that requires either hardware attestation (Section 8.6) or, for the LTX video path, the weight-binding work on the roadmap.

The guarantee that the committed model *actually ran* comes from confidential computing, not from the zkVM: an enclave measures the binary, binds the model by SHA-256 before loading it, and attests to both. That path has now run on real confidential-computing hardware, an Intel TDX confidential VM with an NVIDIA H200 (September 2026, Section 8.6), and paid testnet sessions from the Platformless AI app have run on it, but it is not yet the default path sessions take. Until it is, honest hosts are kept honest by commitments, client-side checks, and stake, and this document says so rather than calling a commitment a proof of computation.

### HostEarnings

Accumulates host payments for gas-efficient batch withdrawals.

- 80% gas reduction vs individual transfers
- Hosts withdraw at their convenience
- Supports multiple payment tokens

## 4.3 SDK Architecture

The SDK (`@fabstir/sdk-core`) is the primary interface for client applications. It is organised into focused modules — one per capability (authentication, payments, sessions, host discovery, storage,

and so on) — all reached through a single SDK object:

```
// One module per capability, reached via the SDK object
const sdk = new FabstirSDKCore({
  /* config */
});
await sdk.authenticate("privatekey", { privateKey });

// Core capability modules
const auth = sdk.getAuthManager(); // Authentication
const payments = sdk.getPaymentManager(); // Deposits/withdrawals
const sessions = sdk.getSessionManager(); // Session lifecycle
const hosts = sdk.getHostManager(); // Host discovery
const storage = sdk.getStorageManager(); // S5 storage

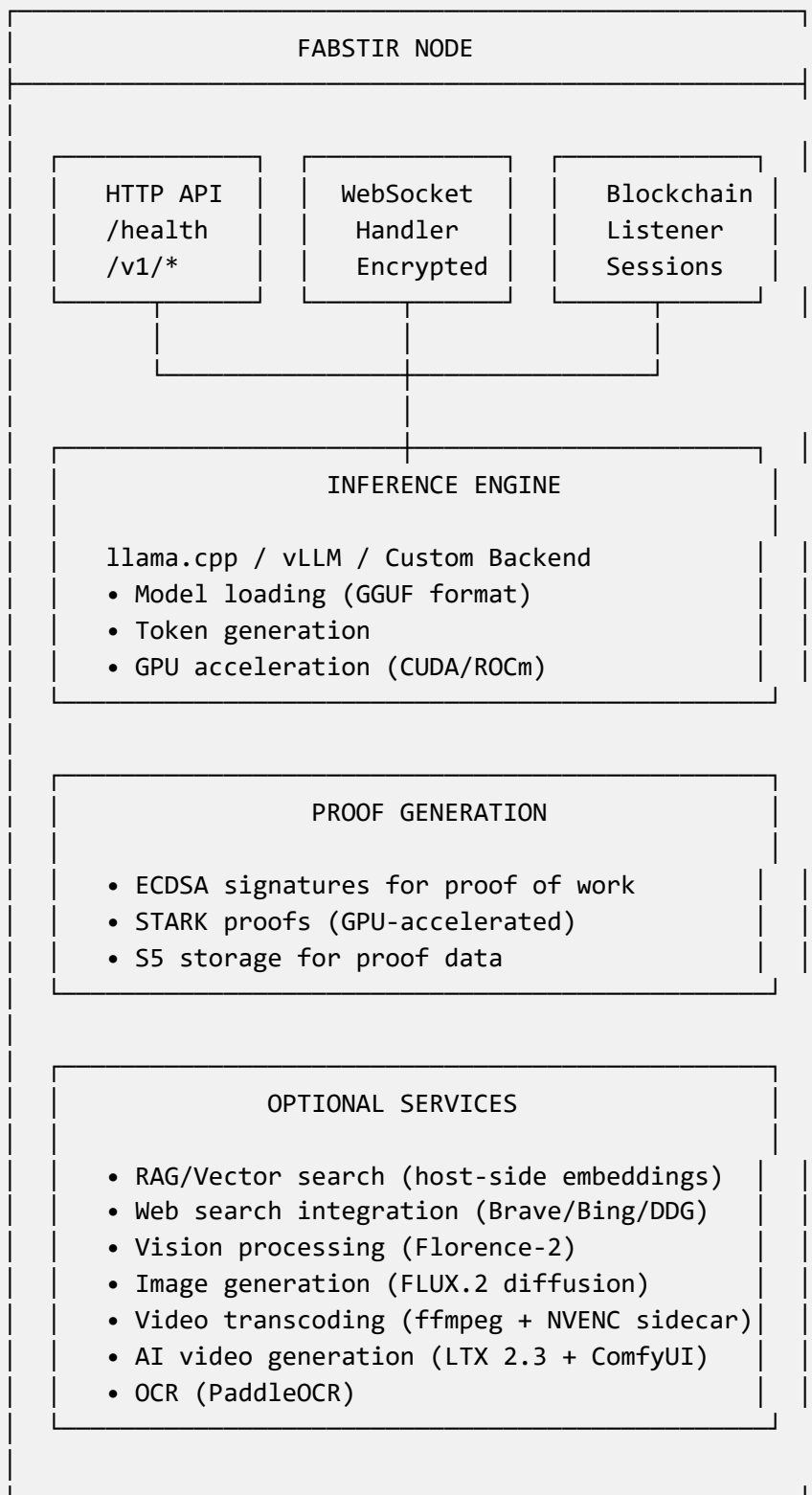
// Advanced capability modules
const vectorSearch = sdk.getVectorRAGManager(); // Vector search (RAG)
const models = sdk.getModelManager(); // Model queries
const transcoding = sdk.getTranscodeManager(); // Video/audio transcoding
```

Key SDK features:

- **Browser-compatible:** Zero Node.js dependencies
- **Multi-chain:** Base Sepolia (production), opBNB Testnet (development)
- **Encryption by default:** All sessions encrypted automatically
- **Deterministic S5 seed:** Derived from wallet signature for data sovereignty

## 4.4 Host Node Architecture

The host node (**fabstir-llm-node**) is written in Rust for performance and security. Rust's ownership model and compile-time memory safety guarantees eliminate entire classes of vulnerabilities (buffer overflows, use-after-free, data races) that plague C/C++ inference servers, making it ideal for handling encrypted user data and financial transactions.



## 4.5 Decentralised Storage (S5)

S5 provides decentralised, content-addressed storage backed by the Sia network.

### Why S5?

- **Content addressing:** Data identified by cryptographic hash (CID)
- **Immutability:** Once stored, content cannot be modified
- **Redundancy:** Data distributed across multiple Sia hosts

- **Cost efficiency:** Storage costs fraction of centralised alternatives
- **Path-based API:** Familiar filesystem interface via Enhanced S5.js

### What We Store on S5:

| Data Type                | Purpose                                           | Encryption                     |
|--------------------------|---------------------------------------------------|--------------------------------|
| Conversation history     | User's chat records                               | User-controlled                |
| Proof data<br>(proofCID) | Complete proof for verification                   | Public                         |
| Delta CIDs               | Incremental proof changes                         | Public                         |
| Vector databases         | RAG document embeddings                           | User-controlled                |
| Source videos            | Original video content                            | User-controlled                |
| Transcoded outputs       | Whole-file video (Phase 1)                        | User-controlled                |
| Generated video<br>clips | LTX video output (all thirteen modes)             | User-controlled                |
| Video input stills       | i2v/flf2v/iclora conditioning frames              | Key-in-CID (u-prefix)          |
| Control clips            | Motion/content source for the guided<br>modes     | Key-in-CID (u-prefix)          |
| Clip archive             | User's own encrypted backup of delivered<br>clips | User-controlled<br>(encrypted) |
| LTX attestations         | Input commitment + frame Merkle                   | Public (keyless)               |
| LTX allow-list bundle    | Host templates + param bounds                     | Public                         |
| HLS init segments        | fMP4 initialisation segment                       | Public (z-prefix)              |
| HLS preview<br>segments  | Free preview fMP4 segments                        | Public (z-prefix)              |
| HLS paid segments        | Encrypted fMP4 segments                           | Key-in-CID (u-prefix)          |
| HLS content<br>metadata  | Segment CIDs, resolutions, durations              | User-controlled                |
| Transcode proof<br>trees | GOP-level Merkle proof trees                      | Public                         |
| Session metadata         | Audit trail                                       | Host-signed                    |

### Data Sovereignty:

Each user's S5 identity is derived deterministically from their Ethereum wallet signature. This means:

- Users own their data (not the platform)
- Data follows the user across devices
- No account recovery needed—wallet = identity

- Data cannot be accessed without the wallet

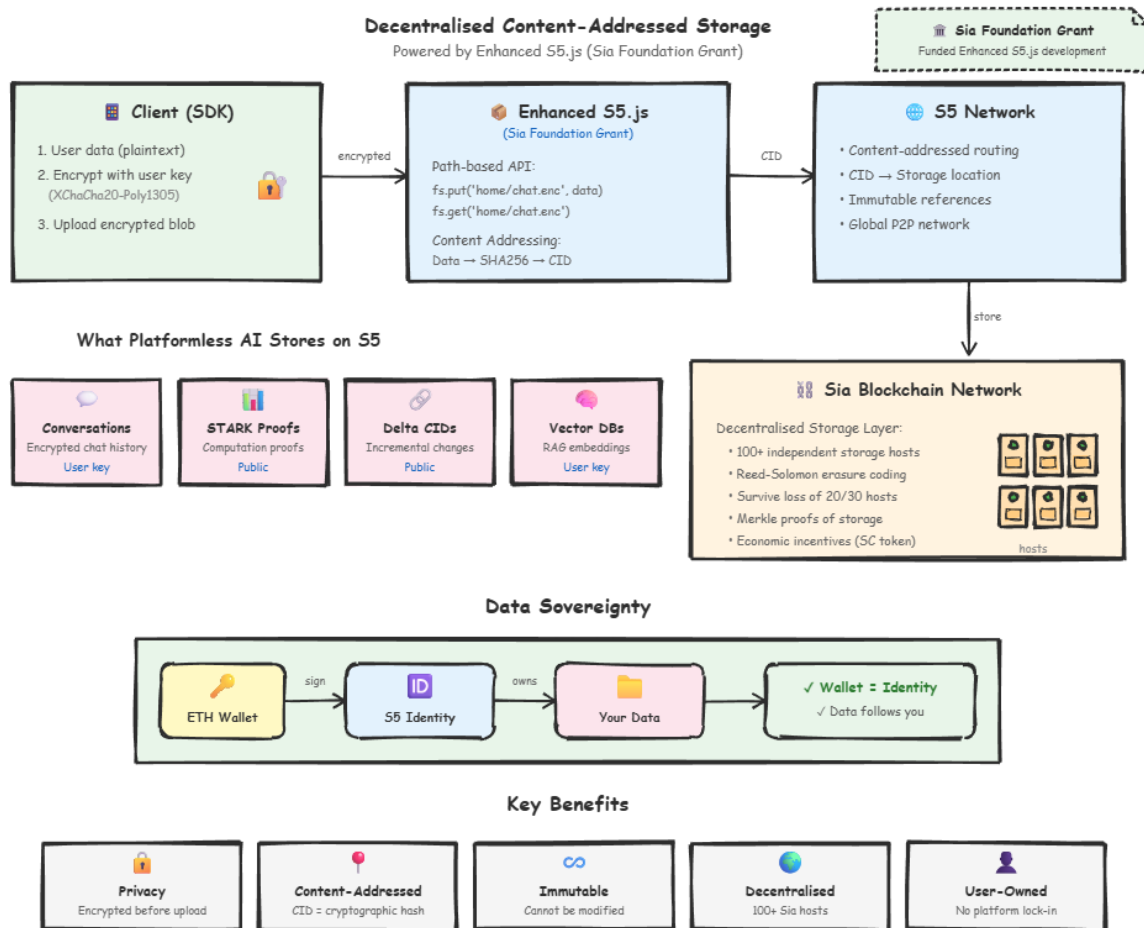


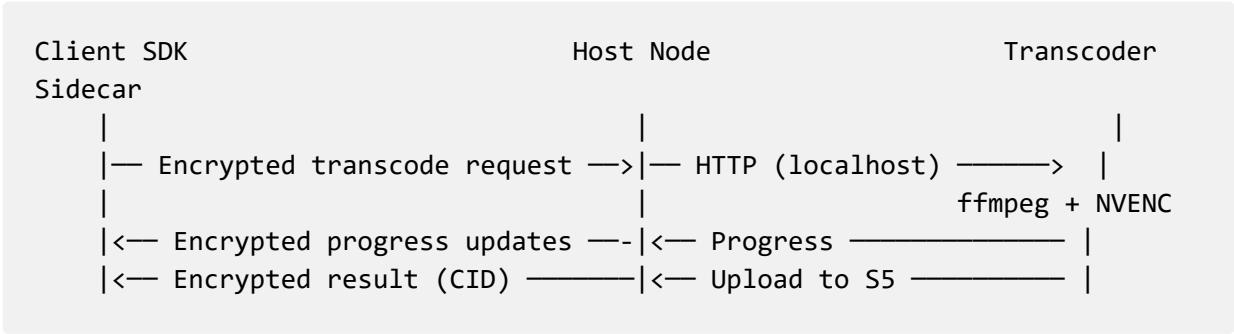
Figure 4: S5 Storage Architecture showing the decentralised content-addressed storage flow from client encryption through Enhanced S5.js to the Sia blockchain network. Developed with Sia Foundation grant funding.

## 4.6 Decentralised Video Transcoding

The protocol extends beyond text-based AI to include **GPU-accelerated video transcoding** — a compute-intensive workload that benefits from the same decentralised, encrypted, and verifiable infrastructure used for inference.

### Architecture

Each host runs a **transcoder sidecar** — an independent Docker container running ffmpeg with NVIDIA NVENC hardware encoding — alongside its main host node. The sidecar communicates with the host node over localhost HTTP, while all client-facing communication flows through the encrypted WebSocket channel.



NVIDIA's NVENC is dedicated encoder circuitry, separate from the CUDA cores used for AI inference. This means a host running LLM inference can simultaneously transcode video without either workload degrading the other — enabling dual revenue streams from the same GPU hardware.

End-to-End Encrypted Video Pipeline

The transcoding pipeline maintains full encryption throughout:

1. **Source upload:** Client encrypts video with XChaCha20-Poly1305 and uploads to S5. The decryption key is embedded in the CID — only the CID holder can decrypt.
2. **Encrypted transit:** The transcode request (source CID, format parameters, job ID) is sent over the E2E encrypted WebSocket channel. Each transcode job establishes its own WebSocket connection with a unique session key via ECDH, enabling multiple concurrent jobs without session interference.
3. **Host-side processing:** The sidecar downloads the encrypted source from S5, decrypts in memory, transcodes via GPU, and encrypts the output before uploading back to S5. Unencrypted video exists only in memory during processing — never persisted to disk.
4. **Result delivery:** The encrypted output CID is returned to the client over the encrypted WebSocket. Only the client can decrypt the transcoded video.

No centralised server ever sees the unencrypted video content.

Capacity-Aware Load Balancing

When multiple hosts support the same transcode model, the SDK's load balancer distributes jobs across them:

1. **Rank:** Hosts scored by blockchain-derived factors (stake, price, uptime, latency) using the configured selection mode (AUTO, CHEAPEST, RELIABLE, FASTEST).
2. **Filter:** Each host's real-time capacity checked via `GET /v1/transcode/capacity`. The SDK also tracks its own pending submissions per host, so rapid sequential submissions don't all land on the same host before the capacity endpoint updates.
3. **Overflow:** When the preferred host's effective capacity reaches zero, jobs automatically overflow to the next-ranked host. If all hosts are temporarily exhausted, the load balancer retries after a configurable backoff.

This was validated in a stress test with 7 concurrent jobs across two GPU hosts (NVIDIA L40S and RTX Pro 6000 Max-Q), achieving 100% completion with automatic overflow when the preferred host's 3 NVENC slots were saturated.

## GOP-Level Verification

Transcoded output is verified at the **Group of Pictures (GOP)** level using the same STARK proof infrastructure used for inference:

- Each GOP produces: input hash (keccak256), output hash, quality metrics (PSNR in dB, SSIM), and a STARK-anchored commitment to those values (as with inference, this commits the host to what it did; it does not re-execute the transcode inside the zkVM)
- GOPs are assembled into a Merkle tree; the root hash is committed on-chain (32 bytes)
- The full proof tree is stored on S5 and can be independently verified by anyone
- Random spot-checks verify individual GOPs against quality thresholds (standard: PSNR >= 38.0 dB)
- On-chain dispute resolution with refund logic if quality violations are demonstrated

## Supported Formats

| Codec      | Encoder    | Status     |
|------------|------------|------------|
| H.264      | h264_nvenc | Production |
| HEVC/H.265 | hevc_nvenc | Production |
| AV1        | av1_nvenc  | Production |

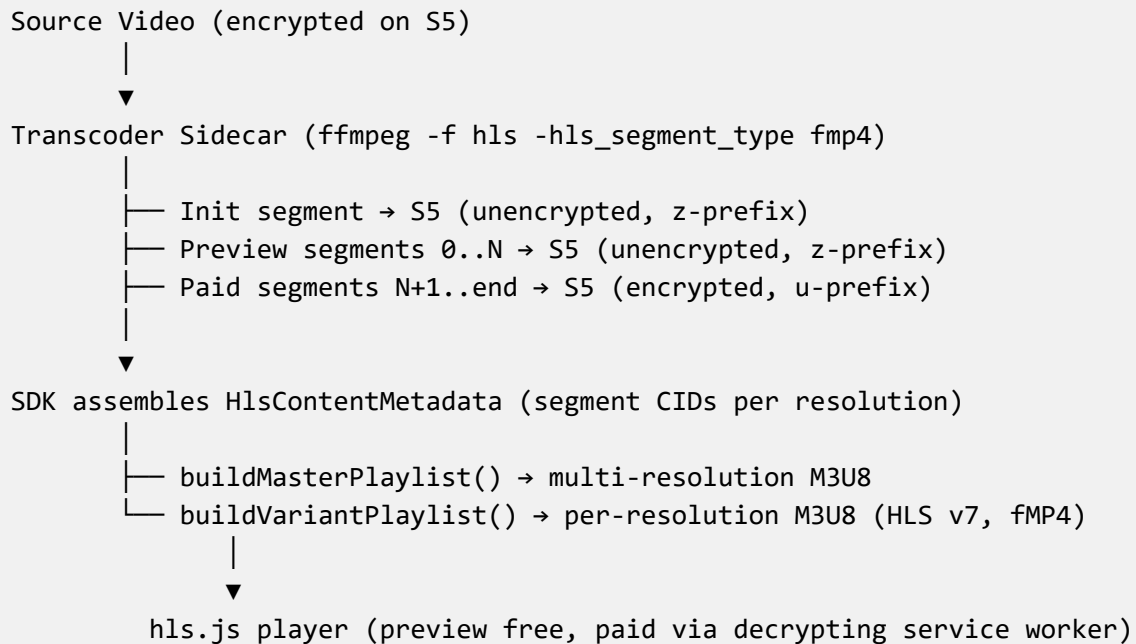
| Output Mode            | Description                                                 | Status     |
|------------------------|-------------------------------------------------------------|------------|
| Whole-file (Phase 1)   | Single CID per format, preview via <b>trim_percent</b>      | Production |
| HLS segments (Phase 2) | fMP4 segments per format, preview via <b>previewPercent</b> | Production |

Multiple output resolutions can be produced in a single job (e.g. 720p + 1080p + 4K).

## HLS Adaptive Bitrate Streaming

The protocol supports **HLS adaptive bitrate streaming** with per-segment encryption — enabling white-label streaming platforms where creators upload content, viewers get a free preview, and paid content is cryptographically gated.

When a format includes **hls: true**, the transcoder sidecar produces **fMP4 segments** instead of a single file. Each segment is uploaded individually to S5, with preview segments unencrypted (**z**-prefix CID, publicly accessible) and paid segments encrypted (**u**-prefix CID, decryption key embedded in CID). The SDK then generates **M3U8 playlists** client-side from the segment metadata.



This pipeline has been **validated end-to-end**: encrypted source upload to S5, GPU transcoding on NVIDIA L40S (480p/720p/1080p AV1), HLS fmp4 segments with XChaCha20-Poly1305 per-segment encryption stored on decentralised Sia/S5, USDC payment settlement, client-side M3U8 playlist generation, and adaptive bitrate playback via hls.js with a decrypting service worker.

## 4.7 Decentralised AI Video Generation (LTX 2.3)

Beyond transcoding video a user already holds, the protocol also **generates video** — running the LTX 2.3 diffusion model on the same encrypted, verifiable, pay-per-use infrastructure. **Thirteen generation modes** are live, each selected by its own pinned template and its own on-chain model id.

Three are generative from scratch or from stills:

- **text-to-video** — a clip from a prompt, seed, and resolution.
- **image-to-video** — the user's own still, encrypted client-side, becomes the first frame and the model animates it.
- **first-last-frame** — two ordered stills; the model synthesises the motion between them.

One conditions on a reference image rather than a clip:

- **ingredients** — a single reference sheet drives a newly generated clip, so recurring subjects, props or looks carry across shots without re-describing them in the prompt each time.

Nine are *guided*: they take a control clip the user already has, and run on the IC-LoRA union-control path, where the node derives the control geometry from the clip in-graph, so the user supplies plain video and never has to produce depth passes or masks:

- **restyle** — a reference still plus a control video whose motion and camera the output follows, re-skinned to the reference look.
- **outpaint** — the source is fitted into a *different* canvas (16:9 into 9:16 for social, or the reverse) and the model paints everything left empty. Where the source sits inside the new

frame is the user's choice, and it decides what the model must invent: a figure cropped at the waist is anchored low so the model paints sky and buildings above it rather than inventing legs.

- **edit** — a prompt-driven change to an existing clip (*add, remove, replace, restyle*), with the rest of the frame preserved.
- **restore** — a degraded clip re-rendered with detail recovered, keeping the original framing.
- **upscale** — a true 2x enlargement, separate from restore because the sampler schedule that enlarges without repainting is not the one that repairs.
- **water** — water added to an existing scene (rivers, rain, surf, floods, splashes), driven by a structured prompt that states what the reference shows and what the edited clip should show.
- **day-to-night** — a daylight clip converted to night, with lighting, sky and practical sources re-derived rather than filtered.
- **cross-view** — novel view synthesis: the scene re-rendered from a camera position the original never occupied, with azimuth, elevation and distance requested within a trained envelope.
- **SDR-to-HDR** — a standard-dynamic-range clip reconstructed as HDR and delivered scene-linear, for grading rather than for display as-is.

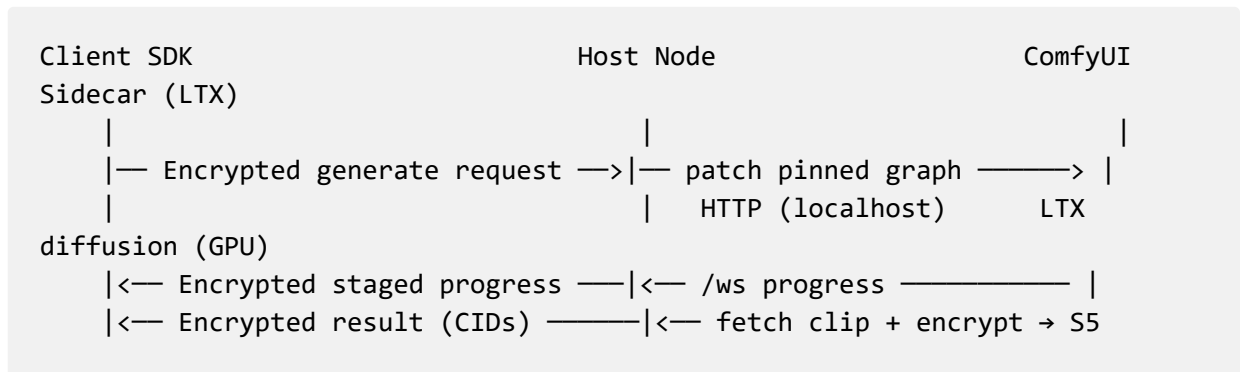
**Delivery.** Every mode can return either an H.264 clip or, opt-in, a **16-bit EXR master sequence** alongside an MP4 preview — half-float, ZIP-compressed, linearised on write, with the frame count of the sequence required to equal the frame count billed, so a short delivery fails closed rather than quietly shipping. The manifest records the colour encoding it actually wrote, which differs by mode: the SDR-to-HDR path is scene-linear with values above 1.0, because reconstructing that range is the point, while every other template delivers linearised display-referred content. That distinction is stated rather than assumed, so a grader can trust the label.

Resolutions span SD through 4K (portrait, landscape, and square), with user-selectable clip length (whole seconds, 5 to 15) and frame rate (24, 25, 48, or 50 fps), including generated audio. All thirteen modes carry the same provenance and settle through the same on-chain escrow; the only difference between them is which pinned template runs and which inputs are bound into the commitment.

**Precision.** The weights are **quantised**, and this document says so rather than letting "the LTX 2.3 model" imply reference precision: the 22B transformer checkpoint carries mixed-precision quantisation metadata (FP8 tensor types, some operations emulated on current hardware), and the Gemma-3 12B text encoder runs at FP4-mixed. Quantisation is what makes a five-second 4K clip an eight-minute job rather than an hour-long one, and it is the honest description of what a buyer receives. It also names a real problem for a *marketplace*, as opposed to a single operator: because billing is metered in frames and pixels rather than GPU seconds, a host that quietly served a more aggressively quantised model would render faster, spend less, charge the same, and deliver a worse clip. Binding the exact weight files by hash — so that "which model ran" means the bytes and not the name — is therefore a prerequisite for opening LTX generation to third-party hosts, and is tracked as such on the roadmap.

## Architecture

Each host runs a **ComfyUI sidecar** — an independent Docker container running ComfyUI with the LTX 2.3 model — alongside its main node, mirroring the transcoder pattern. The sidecar talks to the node over localhost HTTP; all client-facing traffic stays on the encrypted WebSocket channel. Generation runs on the CUDA cores, so a host dedicates the GPU to a clip while it renders, with per-host capacity advertised and rate-limited.



### Pinned-Graph Provenance

The novel problem in decentralised generation is proving *which model and pipeline actually ran*. The protocol solves this with a **pinned, hash-committed ComfyUI graph** and a **versioned allow-list bundle**:

- Each host publishes an **allow-list bundle** (the templates it serves plus parameter bounds — frame counts, frame rates, resolutions) to S5. The client **fetches and authenticates** this bundle against its on-chain **bundleHash** and **allowListVersion** *before paying*, so it knows exactly what it is buying.
- Per job, the node fills the pinned graph by **value substitution only** — writing the prompt, seed, and resolution into the graph's own named nodes, never adding, removing, or rewiring anything. The graph that runs is therefore the graph that was hashed (**templateHash**), preventing a host from silently swapping in a cheaper or different model.
- For image- and video-conditioned modes, the **byte-exact content of every input is bound into the input commitment** (a keccak hash per still and per control video, order-significant — swapping first and last frames changes the commitment). The commitment is versioned by input kind, so a text, image, or video job each commits to exactly the inputs it carried. The client can later prove not just *which model* ran but *exactly which images and which control clip* conditioned the output.

### End-to-End Encrypted Pipeline

1. **Pre-flight**: The client validates the job against the authenticated bundle bounds (templates, frame/fps/resolution ranges, image counts, seed range — a bad job never locks funds), estimates the cost, and escrows USDC on-chain — a dedicated session per clip, sized as **max(floor, exact cost)**. Input stills, if any, are encrypted client-side and uploaded to S5 as private capability CIDs; their decryption keys travel only inside the encrypted channel.
2. **Generation**: Over the encrypted WebSocket, the node patches the pinned graph, drives ComfyUI, and streams **staged progress** (**generating** → **encrypting** → **uploading** → **finalising**).
3. **Encrypted delivery**: The rendered clip is pulled from ComfyUI over localhost HTTP, encrypted with XChaCha20-Poly1305 under per-frame keys, and uploaded to S5.

Unencrypted output exists only in memory; only the client can decrypt the result.

4. **Provenance**: The node returns a **keyless manifest** (per-frame content hashes assembled into a Merkle root) and a **fixed-field attestation** binding the exact inputs — prompt, seed, dimensions, model id, template hash, and the content hashes of any input images — to the output CID. The client verifies this **input-binding**, confirming the delivered clip is the one it paid for, from the model it selected, conditioned on exactly the images it supplied.
5. **Settlement with proof anchoring**: The host submits its proof of work on-chain — the SHA-256 of the stored attestation plus the tokens claimed — then completes the session after the dispute window. The contract pays the host and refunds the remainder automatically, and the client independently checks that the on-chain proof hash matches the attestation bytes it holds (**integrity: ok**). The attestation is thereby **anchored**: a third party can later fetch it from S5, hash it, and compare against the chain.

## Billing

Generation is metered in **megapixel-frames** ( $\text{frames} \times \text{width} \times \text{height}$ ), priced from the host's on-chain registration — a longer or larger clip costs proportionally more, and the client sees the price, read directly from the chain, before committing. On current test pricing a 5-second clip runs from about \$0.04 (SD) through \$0.23 (1080p) and \$0.40 (1440p) to \$0.91 (4K); settlement splits 90% to the host and 10% to the protocol treasury, with the unused deposit refunding automatically. The client enforces the billing honestly in both directions: it rejects any over-claim beyond its own estimate, and the on-chain tokens claimed must equal the wire billing exactly.

## The Resolution Grid

LTX 2.3 quantises its latent, and a protocol that bills before it renders has to be honest about the consequences rather than discover them at delivery. A requested dimension is delivered rounded **down to the nearest multiple of 64**: 1280×720 arrives as 1280×704, and 3840×2160 arrives as 3840×2112. Any resolution that is already a multiple of 64 in both axes is delivered exactly as asked.

The guided modes carry a **separate** constraint, and it is genuinely a different rule rather than the same one restated. Their IC-LoRA guide halves the latent again, so both latent dimensions must be **even**:  $\text{floor}(w/32)$  and  $\text{floor}(h/32)$  must each be divisible by two. That is not the same test as "a multiple of 64". 1280×720 has a latent height of  $\text{floor}(720/32) = 22$ , which is even, so a guided render succeeds and simply delivers 704 lines; whereas 1080 gives  $\text{floor}(1080/32) = 33$ , which is odd, so **1080p and 1440p are impossible** on any guided mode, and the host would spend the text encode and the depth pass before ComfyUI refused.

So a buyer faces two independent questions: *can this render at all* (an even latent, on the guided modes) and *will it arrive at the size I asked for* (a multiple of 64, on any mode). The allow-list bundle (v25 at the time of writing) carries the rungs that satisfy both, including **1920×1088** and **2560×1408** in place of 1080p and 1440p, and the client and the host each refuse an unrenderable shape **before any money moves**.

## Validated

The full pipeline — generation, encryption, delivery, provenance, **and settled economics** — has been **proven end-to-end on Base Sepolia with a live RTX PRO 6000 host**, across the mode set

and from three independent clients (the SDK's scripted gates, the Platformless AI product UI, and the Blender extension). Paid sessions have settled on-chain with host earnings, treasury fee, and user refund each verified **to the unit** against the contract's settlement formulas; both settlement paths (immediate and dispute-window-gated) have been exercised live; and the product UI displays a provenance badge the user's own browser computed, upgrading to "anchored" once the host's proof confirms on-chain.

**4K is now live-proven**, not staged: a 5-second 3840×2160 text-to-video job settled at 1,003,623 billed tokens after 8 minutes 15 seconds on the RTX PRO 6000, billed exactly to its pre-flight estimate. Because cost scales with tokens rather than pixels — resolution, duration, and frame rate all multiply — the client caps a single job by token count (2,000,000, twice the proven 4K point) rather than by resolution, so a job is refused for the thing that actually makes it expensive. Next milestones: host-signed attestations, and C2PA/Content Credentials export backed by the on-chain anchor.

### In the Filmmaker's Timeline (Blender)

The same paid, provenance-bound generation now runs **inside Blender**, across **all thirteen modes**. A native Blender 5.x extension places generation directly in the Video Sequence Editor, and the gesture is always the timeline itself: a prompt for text-to-video; a keyframe still for image-to-video; two stills to bridge for first-last-frame, where **the gap between them is the timing** (the extension fits the clip to the gap rather than asking the artist to compute a duration); a movie strip plus a reference still to **restyle**; and a single movie strip for **outpaint**, **edit**, or **restore**. An annotated placeholder appears immediately, and the finished clip replaces it **in place**, at the exact frame count and price shown on a cost card before anything is spent.

For the guided modes the extension performs a **conform render**: it renders the selected control window out of the artist's own sequencer at the job's exact frame rate and resolution before upload, so the clip that drives the output is exactly the timeline content that was chosen — including, for outpaint, the artist's chosen anchor within the new canvas — and the delivered clip's audio is carried through as a frame-aligned sound strip above it.

Delivered clips can be **archived to the user's own encrypted Sia account** and restored back onto the timeline later. The clip is encrypted client-side under the user's own S5 seed before it leaves the machine; the archive is not a platform's copy of the artist's work but the artist's own, and a clip deleted from local disk is restored from Sia into the same strip.

The extension is a pure client of the protocol — the host node is unchanged — reaching it through a local helper daemon that wraps the SDK, so every clip carries the same input-bound attestation and settles on-chain to the unit. This has been **proven live on Base Sepolia**: clips generated from within Blender, from SD through 720p and HD, each billed exactly as its estimate, with the host's earnings, the treasury fee, and the user's refund verified against the contract's settlement formulas, and each clip's session id, proof CID, seed, and billing recorded on the delivered strip. Because the LTX 2.3 pipeline is HDR and EXR-native, this places verifiable, privacy-preserving AI video generation inside a professional film and VFX production tool rather than a consumer web app.

On 30 September 2026 the extension rendered from a host deployed in a confidential VM with an NVIDIA H200 (Section 8.6): text-to-video, image-to-video, first-last-frame, edit, upscale and

outpaint, every job settled on-chain, and the text-to-video, image-to-video and outpaint jobs each checked against their pre-flight estimate and billed exactly to it. The video weights on that machine are public and were verified against pinned SHA-256 hashes as they downloaded; unlike the language model's weights, they are not sealed behind the attested key release.

## 4.8 Decentralised Fine-Tuning (Training M0)

Every capability described so far consumes a model. Fine-tuning **produces** one, and that changes what the protocol has to protect. In inference the user's exposure is a prompt that lives for the length of a session. In training the input is the user's own corpus — support transcripts, a house style, a proprietary domain — and the output is a durable asset whose value is precisely that nobody else holds it.

So the design goal is narrower than "train a model on rented hardware": the user's dataset must reach the GPU encrypted, the resulting adapter must come back encrypted, and the host must be paid for exactly the work it did and not a token more, all without either party trusting the other.

**How a job runs.** The client shards its dataset, encrypts each shard under a fresh key, and uploads the shards to S5 as capability CIDs — the same key-in-CID scheme the video inputs use, so possession of the pointer is the authorisation and the storage network never holds a key. The node fetches the shards, verifies each against its manifest hash, re-assembles, scans the content, and re-counts the tokens before any GPU cycle is spent. Training then proceeds in **slices**, and each slice delivers its encrypted checkpoint pointer to the client *before* its proof is submitted on chain. Money never moves ahead of the client holding the artefact. At the end the adapter itself is returned the same way: sharded, encrypted, and pointed to by a manifest whose hash is bound into the final attestation.

**The interesting problem is counting, not training.** The client must declare how many tokens its dataset represents in order to size the escrow, and the node must be able to reject a false declaration — otherwise the escrow is a guess. That means two independent implementations, in two languages, must agree on a token count exactly, for arbitrary user text. A disagreement is not a rounding error; it rejects an honest job. The protocol pins this with a frozen counting fixture generated from the exact **tokenizer.json** bytes the template names, and both sides implement counting strictly from those cases. The failure mode this guards against is real and specific: JavaScript strings are UTF-16 while the reference implementation is UTF-8, so a client that walks code units rather than code points diverges on every character outside the Basic Multilingual Plane — which is to say, on every emoji — and on nothing else.

**Serving the result.** A finished adapter can be attached to an ordinary inference session by pointer. The node fetches it, verifies it against the manifest hash, stages it private to that session, and applies it to that session's requests only. It is never visible to another session on the same base model, and it is removed when the session ends. The registry key an adapter is held under is minted by the node and never accepted from the wire, so naming another user's session buys nothing.

Because the adapter lives on S5 as an encrypted artefact and the capability pointer is the whole grant, it is **portable across hosts by construction**: the host that trained it holds no privileged position over the host that serves it. This is demonstrated, not theoretical — an adapter trained on one operator's GPU has been staged from storage and served on a different operator's machine, a

different hardware generation, with the same per-session isolation. Train anywhere, serve anywhere, keys with the customer throughout.

**On an attested confidential VM.** On 30 September 2026 the same job ran on the confidential-VM host of Section 8.6: a QLoRA fine-tune of Qwen3.8-27B on a fictional corpus, 43,505 billed tokens. On the host's side the dataset was decrypted only inside the confidential VM; the adapter was trained on the H200, returned encrypted, and then served back into an ordinary session on the same machine, where it answered the corpus's invented facts that the base model, asked the same questions, could not. Two independent runs of the same job on that machine produced **byte-identical adapters**, assembled from different encrypted uploads: the training is reproducible to the bit, which makes a result checkable by running it again.

**Settlement is per slice, not per job.** A run that dies halfway bills for the slices that completed and nothing more, and a client that cancels is billed the same way. A job rejected before any GPU work — a token count that does not reconcile, a dataset that fails its content scan, a busy host — settles to zero and frees the deposit rather than stranding it.

**It required no new smart contracts.** Fine-tuning rides the same session-job machinery as inference: the same escrow, the same per-slice proof submission, the same dispute window and 90/10 settlement split, distinguished only by its own registered model id. A training job is, to the chain, a session that consumes tokens and submits proofs — which is what makes the economics above inherited rather than invented, and what let the capability be added without touching a deployed contract.

The same rails extend beyond text. Because a slice is defined by a token budget rather than by anything language-specific, video LoRA training sits on the identical accept, slice, settle and delivery path, with the template and the counting recipe carrying the difference.

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## 5. Agentic AI and SaaS Infrastructure

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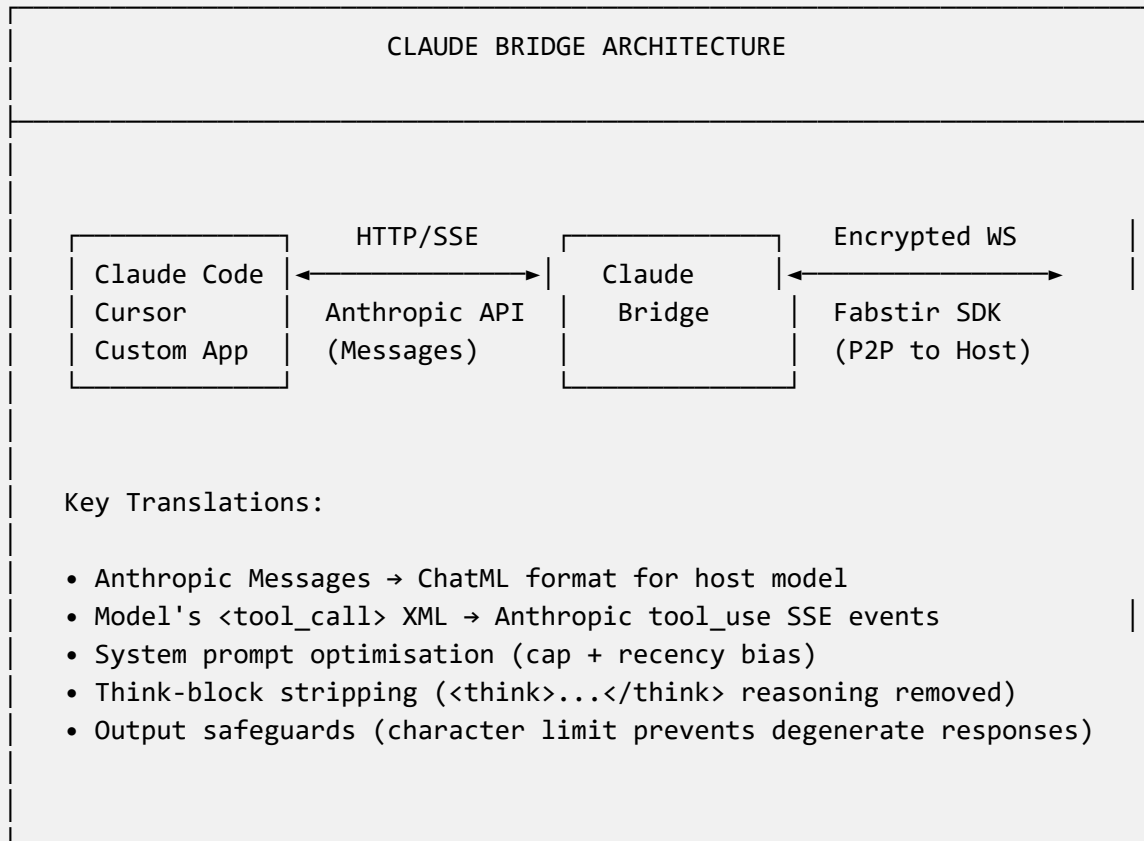
### 5.1 Beyond Chat: The Agentic AI Opportunity

Traditional AI inference is request-response: a user sends a prompt, receives a response. **Agentic AI** is fundamentally different — autonomous agents that plan, use tools, observe results, and iterate. These agents may execute dozens of tool calls per session: reading files, running terminal commands, editing code, searching the web — all without human intervention between steps.

This represents a massive market shift. Agentic coding tools (GitHub Copilot, Cursor, Claude Code, Windsurf) are growing rapidly, with the agentic AI market projected to exceed \$15B by 2027. Every one of these tools currently depends on centralised API providers — creating the same vendor lock-in, privacy, and censorship risks described in Section 2.

### 5.2 Claude Bridge: Anthropic API Compatibility Layer

The **Claude Bridge** ([@fabstir/claude-bridge](#)) is an API translation layer that enables any Anthropic Messages API-compatible client to use Platformless AI's decentralised infrastructure without modification.

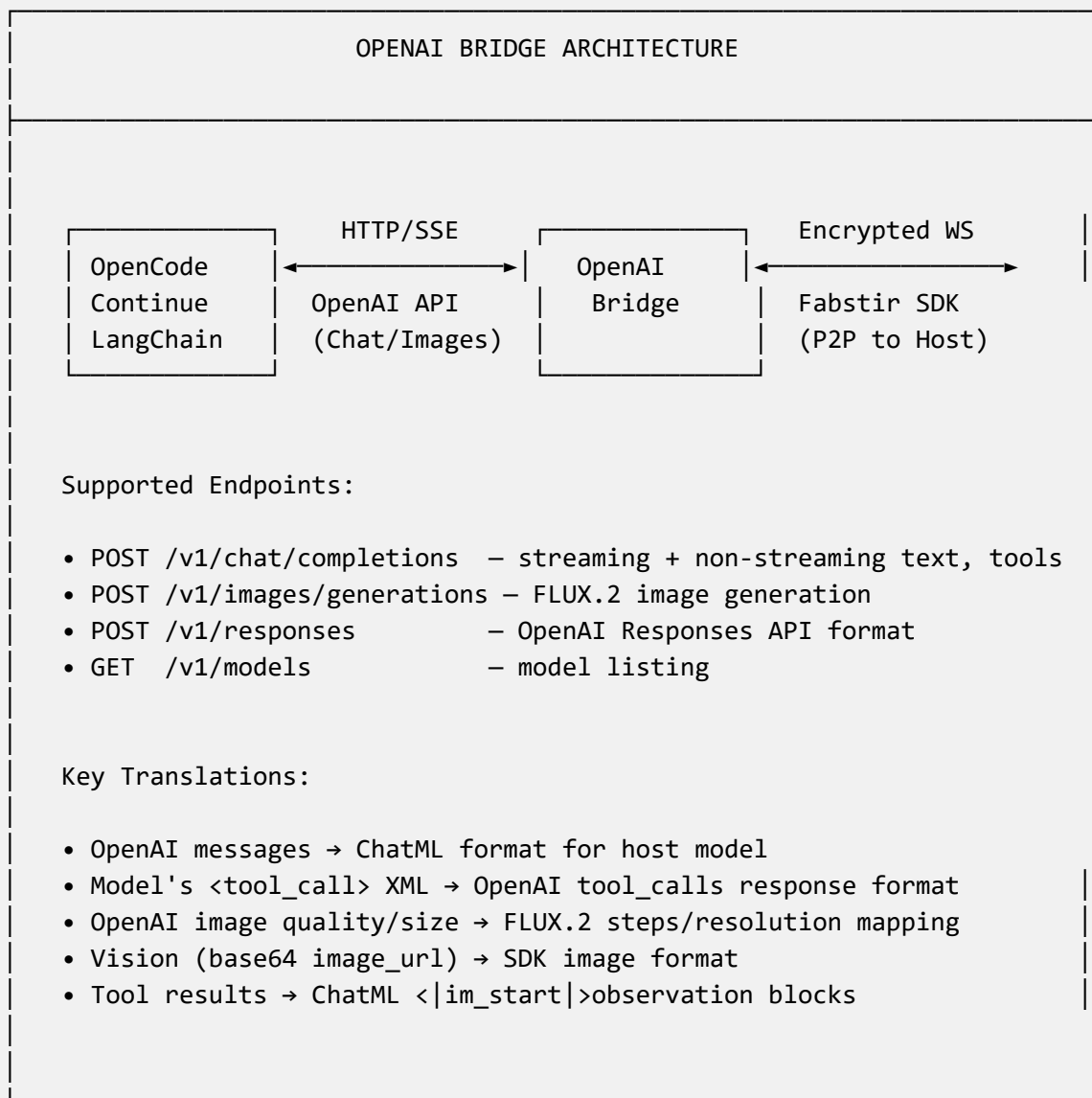


#### Technical implementation:

- **SSE streaming:** Full Anthropic streaming protocol — `message_start`, `content_block_start`, `content_block_delta`, `content_block_stop`, `message_delta`, `message_stop`
- **Tool use:** `tool_use` content blocks with `input_json_delta` events, proper `stop_reason: "tool_use"`
- **Multi-turn tool results:** `tool_result` messages serialised into ChatML `<|im_start|>observation` blocks
- **Session management:** Automatic session creation, encrypted WebSocket pooling, session auto-recovery on disconnect
- **Output safeguards:** Character limit based on `max_tokens` prevents degenerate model responses from consuming GPU indefinitely

### 5.3 OpenAI Bridge: OpenAI API Compatibility Layer

The **OpenAI Bridge** ([@fabstir/openai-bridge](#)) extends the same principle to the OpenAI ecosystem. Any application using the OpenAI SDK — including OpenCode, Continue, LangChain, and custom applications — can use Platformless AI's decentralised infrastructure by changing a single environment variable.



**Integration is minimal** — applications using the OpenAI SDK simply set their base URL:

```
# Start the bridge (handles blockchain, encryption, sessions automatically)
npx fabstir-openai-bridge --private-key $KEY --model "repo:file"
```

```
# Client configuration
```

```
export OPENAI_BASE_URL=http://localhost:3457/v1
```

```
import OpenAI from "openai";

const client = new OpenAI({
  baseURL: "http://localhost:3457/v1", // OpenAI Bridge endpoint
  apiKey: "any-key", // Bridge handles auth via wallet
});

// Text inference – works unchanged
const response = await client.chat.completions.create({
  model: "any-model", // Mapped to host model
  messages: [{ role: "user", content: "Hello" }],
  stream: true,
});

// Image generation – maps to FLUX.2 diffusion sidecar
const image = await client.images.generate({
  prompt: "A futuristic cityscape at sunset",
  size: "1024x1024",
  quality: "hd", // Maps to 20 inference steps
});
```

The OpenAI Bridge handles blockchain session creation, host discovery, E2E encryption, and session re-keying automatically. The first request triggers lazy initialisation; subsequent requests reuse the session.

## 5.4 Production Validation

Both bridges have been validated in production. The Claude Bridge has been tested with Claude Code — Anthropic's autonomous coding agent that uses 23 tools. The OpenAI Bridge has been tested with OpenCode running against decentralised hosts, including image generation via FLUX.2 diffusion. The chat model a host serves is its own choice and changes over time; at the time of writing the reference deployment serves Qwen3.8-27B at 8-bit, a 27B dense model with a 262k-token native context window.

**Claude Code (via Claude Bridge)** uses 23 tools:

- **File operations:** Read, Edit, Write, Glob, Grep
- **Terminal:** Bash command execution
- **Web:** WebFetch, WebSearch
- **Planning:** EnterPlanMode, ExitPlanMode, Task delegation
- **Notebook:** NotebookEdit for Jupyter notebooks

**Demonstrated capabilities:**

- Creating a complete React application from scratch (scaffolding, editing, starting dev server)
- Multi-turn tool conversations (read file → edit file → verify changes)
- Background task management (dev servers running while agent continues working)
- Error recovery (agent adapts when tool calls fail)

All tool calls flow through encrypted P2P WebSocket connections to decentralised GPU hosts. The source code, prompts, and responses never touch a centralised platform.

5.5 SaaS AI Backend

For companies building AI-powered products, the Claude Bridge provides a **drop-in replacement for the Anthropic API** and the OpenAI Bridge provides a **drop-in replacement for the OpenAI API**:

| Feature           | Anthropic / OpenAI API  | Platformless AI                              |
|-------------------|-------------------------|----------------------------------------------|
| API compatibility | Native                  | Full Messages API + Chat Completions API     |
| Data privacy      | Trust platform          | End-to-end encrypted, no intermediary        |
| Vendor lock-in    | Single provider         | Any host, any model                          |
| Rate limits       | Platform-imposed        | No artificial limits                         |
| Pricing           | Platform markup         | Direct GPU market rates                      |
| Censorship        | Platform policies       | Permissionless                               |
| Compliance        | Trust platform          | Cryptographic guarantees (GDPR, HIPAA ready) |
| Availability      | Single point of failure | Distributed, multi-host failover             |
| Image generation  | Platform-hosted         | Decentralised FLUX.2 via OpenAI Bridge       |

**Integration is minimal** — SaaS applications using the Anthropic or OpenAI SDK simply point their base URL to the bridge endpoint:

```
import Anthropic from "@anthropic-ai/sdk";

const client = new Anthropic({
  baseURL: "http://bridge.example.com/v1", // Claude Bridge endpoint
  apiKey: "any-key", // Bridge handles auth via wallet
});

// Everything else works unchanged
const response = await client.messages.create({
  model: "claude-3-opus-20240229", // Mapped to host model
  max_tokens: 1024,
  messages: [{ role: "user", content: "Hello" }],
});
```

**OpenAI SDK integration:**

```
import OpenAI from "openai";

const client = new OpenAI({
  baseURL: "http://bridge.example.com/v1", // OpenAI Bridge endpoint
  apiKey: "any-key", // Bridge handles auth via wallet
});

// Chat – works unchanged
const response = await client.chat.completions.create({
  model: "any-model",
  messages: [{ role: "user", content: "Hello" }],
});

// Image generation – works unchanged
const image = await client.images.generate({
  prompt: "A cat astronaut",
  size: "1024x1024",
  quality: "hd",
});
```

## 5.6 Market Implications

The agentic AI and SaaS infrastructure capabilities — with both Anthropic and OpenAI API compatibility — position Platformless AI to capture value from three rapidly growing markets:

1. **Agentic Coding** (\$15B+ by 2027): Enterprise development teams using AI coding assistants need IP protection. Source code sent to centralised APIs is a liability. Platformless AI provides the same capability with end-to-end encryption, supporting both Anthropic-compatible tools (Claude Code, Cursor) and OpenAI-compatible tools (OpenCode, Continue, LangChain).
2. **AI-Powered SaaS** (\$30B+ by 2027): Every SaaS company embedding AI into their product needs reliable, private inference infrastructure. A single API key dependency is a business risk. Platformless AI eliminates this with decentralised, multi-host infrastructure and drop-in compatibility for both major API standards.
3. **AI Agent Orchestration** (emerging): Multi-agent workflows — research agents, planning agents, implementation agents working in teams — require sustained, tool-heavy sessions. Platformless AI's orchestrator with A2A protocol support, DAG-based task decomposition, and x402 inter-agent payments is purpose-built for this market.

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## 6. Multi-Agent Orchestration and A2A Protocol (Experimental)

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*Note: The A2A protocol was introduced by Google in April 2025 and remains at release candidate status (v1.0.0-rc) as of March 2026. Platformless AI's orchestration layer is functional and fully tested (351 unit tests), but should be considered experimental infrastructure — the underlying standard is ~12 months old and may evolve as it approaches stable 1.0. Our implementation will track the specification as it matures.*

### 6.1 The Need for Multi-Agent Orchestration

Single-model inference is insufficient for complex goals. Analysing a quarterly earnings report requires extracting metrics, comparing historical performance, researching market context, and drafting a summary — tasks best handled by different models with different strengths. The [@fabstir/orchestrator](#) package decomposes complex goals into Directed Acyclic Graphs (DAGs) of sub-tasks, routes each to the optimal model across decentralised GPU hosts, executes them with encrypted inference, settles payments on-chain, and collects cryptographic proofs.

## 6.2 Google's Agent-to-Agent (A2A) Protocol

The [Agent-to-Agent \(A2A\) protocol](#) is an open standard introduced by Google in April 2025 for enabling AI agents to discover, communicate, and collaborate — regardless of framework, vendor, or runtime. Transferred to the Linux Foundation in January 2026, it has 150+ technology partners across enterprise, cloud, and AI ecosystems. The protocol is currently at **v1.0.0-rc** (release candidate) — rapidly approaching stable 1.0 but not yet finalised.

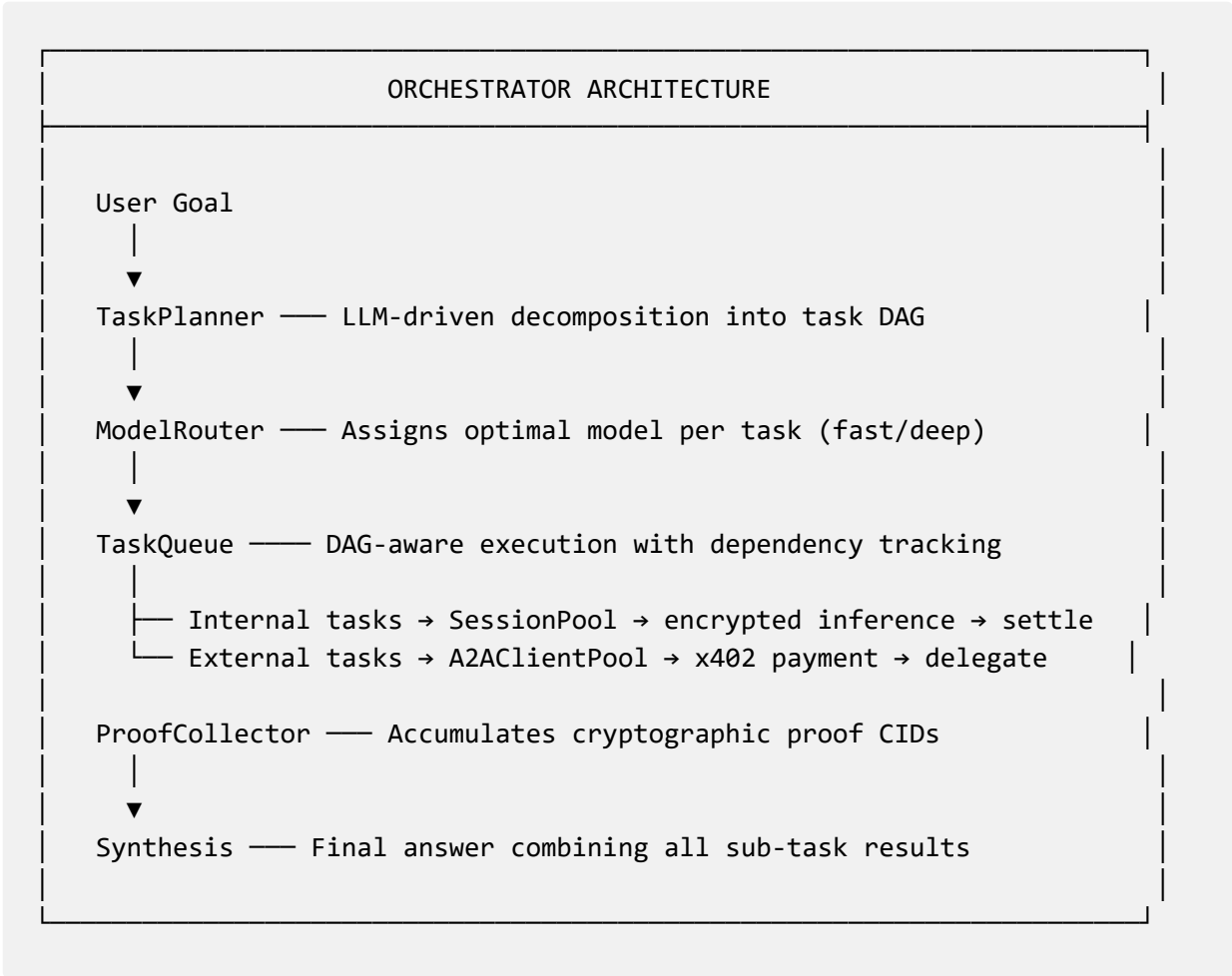
**Platformless AI is, to our knowledge, the first and only DePIN project to implement A2A** — bridging decentralised compute infrastructure with the emerging standard for inter-agent communication. As an early adopter of a pre-1.0 specification, our implementation may require updates as the standard evolves, but the architectural foundations (Agent Cards, JSON-RPC 2.0, SSE streaming) are stable and unlikely to change significantly.

A2A defines:

- **Agent Cards** ([/.well-known/agent.json](#)): JSON metadata describing an agent's capabilities, skills, supported protocols, and authentication — enabling automatic discovery
- **JSON-RPC 2.0**: Standardised request/response format for task delegation between agents
- **Server-Sent Events (SSE)**: Real-time streaming for long-running tasks with progress updates
- **Skill-based routing**: Agents advertise tagged skills; clients discover agents by matching skill tags to requirements

Until now, DePIN AI projects have operated as isolated inference endpoints. A2A transforms this into a collaborative agent network where agents discover each other's capabilities automatically, complex tasks get delegated to the most suitable agent, and results flow back through standardised channels.

## 6.3 Orchestrator Architecture



**Task Decomposition:** The **TaskPlanner** uses a persistent planning session to break complex goals into DAGs of typed sub-tasks (**analysis**, **synthesis**, **research**, **tool-calling**, **creative**), each with optional model preferences.

**Intelligent Model Routing:** The **ModelRouter** assigns models based on task complexity — deep models for tool-calling, synthesis, and large analysis tasks; fast models for research and small analysis — validated against the on-chain **ModelRegistry**.

**Orchestration Patterns:** Three built-in patterns cover common workflows:

| Pattern   | Description                                           | Use Case                                        |
|-----------|-------------------------------------------------------|-------------------------------------------------|
| FanOut    | Execute N sub-tasks in parallel, collect all results  | Independent research queries, parallel analysis |
| Pipeline  | Sequential execution, each task receives prior output | Staged refinement, multi-step reasoning         |
| MapReduce | Parallel map phase, single reduce synthesis step      | Document analysis, data aggregation             |

**Session Pooling and Multiplexing:** The **SessionPool** manages concurrent **FabstirSDKCore** instances with semaphore-based concurrency control, abort signal propagation, and transaction nonce serialisation for safe concurrent on-chain settlement. **Session multiplexing** further reduces costs: when a sub-task finishes, its blockchain session is cached by model rather than destroyed. The next sub-task using the same model reuses the cached session — no new SDK creation, no new

`startSession()`, no additional USDC deposit. For a typical orchestration with 1 planning + 5 same-model sub-tasks, this reduces deposits from 6 to 2 (67% savings). Sessions are only destroyed during `destroy()` at the end of orchestration.

## 6.4 SSE Streaming and Task Lifecycle

The orchestrator supports real-time progress streaming via Server-Sent Events, giving clients granular visibility into every stage of orchestration:

- **Content negotiation:** `POST /v1/orchestrate` with `Accept: text/event-stream` returns an SSE stream; without it, a sync JSON response is returned (fully backward compatible)
- **Phased progress events:** `decomposing` (goal breakdown), `executing` (per sub-task start/completion with task ID), `synthesising` (final answer generation)
- **Artifact streaming:** Synthesis results delivered as `artifact-update` events before the final `completed` status
- **Client disconnect handling:** If the client closes the SSE connection, orchestration is aborted via `AbortSignal` — no wasted compute
- **Task cancellation:** `DELETE /v1/orchestrate/:taskId` cancels an active streaming task

## 6.5 x402 HTTP Payment for A2A Delegation

The x402 HTTP payment protocol enables USDC micropayments for inter-agent delegation, running **alongside** the existing escrow model:

- **Internal inference** (own hosts): Unchanged escrow via `JobMarketplace`
- **External A2A delegation** (foreign agents): x402 per-request USDC micropayment via EIP-3009 `transferWithAuthorization`

**Inbound payments** (server): External agents can pay USDC to use the orchestrator as an alternative to JWT authentication. The server returns HTTP 402 with structured payment requirements; clients respond with a signed `X-PAYMENT` header.

**Outbound payments** (client): When delegating to external agents that return HTTP 402, the orchestrator automatically parses requirements, checks budget, signs an EIP-712 `TransferWithAuthorization`, and retries with the payment header.

Key properties:

- **Gasless USDC transfers:** EIP-3009 — the server submits the signed authorisation on-chain, no pre-funding required from the client
- **Budget enforcement:** `X402BudgetTracker` tracks cumulative outbound x402 spend with configurable limits
- **Agent Card advertising:** x402 pricing advertised in the Agent Card's `x402.accepts` field for automatic discovery
- **Account Kit V2 integration:** Sub-account signer provides scoped USDC spending authority for outbound payments

## 6.6 A2A Server and Client

**Server:** The **OrchestratorA2AServer** exposes the orchestrator as an A2A-compatible agent. It publishes an Agent Card at `/.well-known/agent.json` advertising skills, authentication options (JWT or x402), and pricing. Clients send orchestration requests via JSON-RPC; the server streams progress via SSE.

**Client:** The **A2AClientPool** and **AgentDiscovery** modules allow the orchestrator to discover external A2A agents by skill tags, delegate tasks to them, handle x402 payment negotiation automatically, and cache Agent Cards to avoid repeated network calls.

This means the orchestrator can both **serve** A2A requests from other agents and **delegate** to external A2A-compatible agents — enabling true multi-agent collaboration across organisational boundaries.

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## 7. Economic Model

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### 7.1 Token Economics

The protocol uses two tokens:

#### FAB Token (Native)

- Utility token for the Platformless AI ecosystem
- Used for host staking (minimum 1000 FAB)
- Governance voting rights (used for model approval voting)
- Slashing collateral for dispute enforcement (owner-controlled at MVP, DAO-governed post-MVP)

#### USDC (Stablecoin)

- Primary payment currency for sessions
- Stable value for predictable pricing
- No volatility risk for users or hosts

#### ETH (Native Currency)

- Alternative payment option
- Direct blockchain settlement
- Preferred by crypto-native users

### 7.2 Pricing Model

Hosts set prices per **million tokens** with a precision multiplier of 1000x.

```
Actual USD per million = pricePerToken / 1000
```

Examples:

- `pricePerToken = 60` → \$0.06/million (budget models)
- `pricePerToken = 5000` → \$5.00/million (standard models)
- `pricePerToken = 25000` → \$25.00/million (premium models)

This allows prices from \$0.001/million to \$100,000/million, accommodating everything from tiny test models to premium specialised inference.

### Price Discovery:

- Hosts compete on price for each model
- Users query `getModelPricing()` to compare hosts
- Market forces drive prices towards equilibrium
- No platform markup—users pay hosts directly

## 7.3 Payment Flow

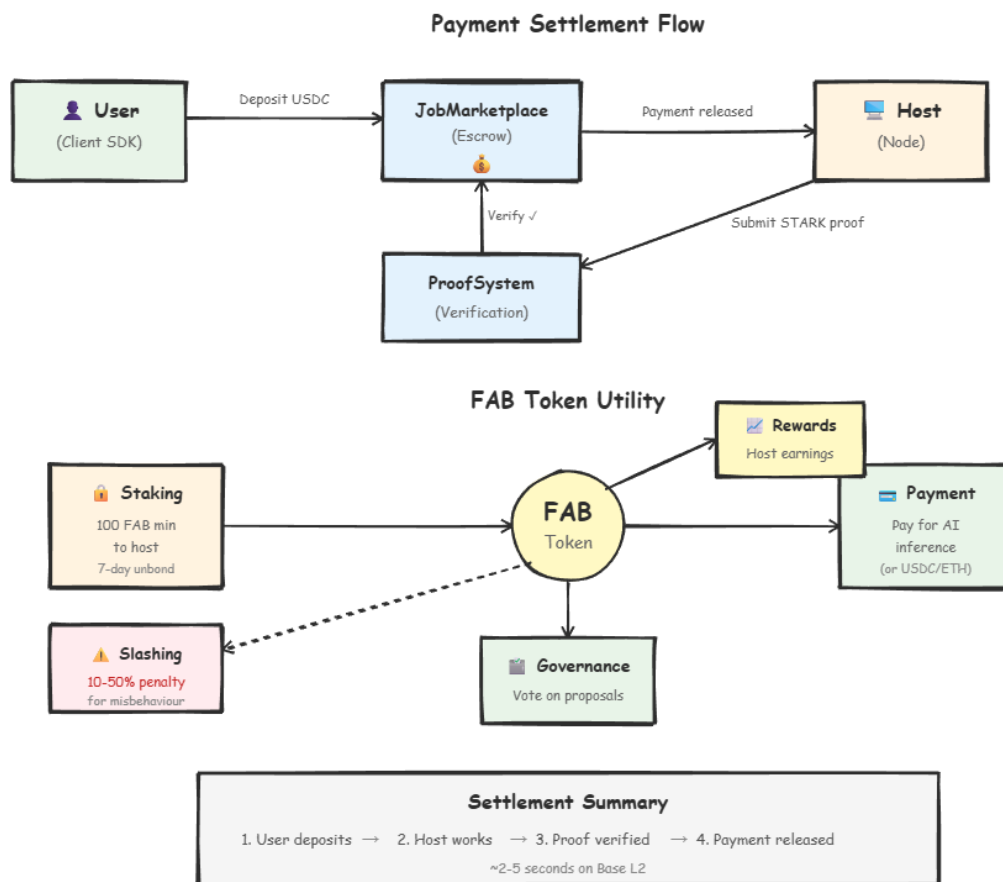


Figure 5: Token Economics & Payment Flow showing the escrow-based settlement process and FAB token utility.

### Fee Structure:

- **90%** → Host (via HostEarnings contract)
- **10%** → Protocol treasury

### Accumulation Pattern:

To save gas, host payments and treasury fees accumulate in contracts rather than being sent immediately. Hosts withdraw when balances are sufficient to justify gas costs. This reduces gas by ~80% compared to individual transfers.

## 7.4 Staking Requirements

| Requirement               | Value        | Purpose                                  |
|---------------------------|--------------|------------------------------------------|
| Minimum host stake        | 1000 FAB     | Sybil resistance, skin in the game       |
| Maximum slash per action  | 50% of stake | Prevents accidental/malicious total loss |
| Minimum stake after slash | 100 FAB      | Below this triggers auto-unregister      |
| Slash cooldown            | 24 hours     | Prevents rapid-fire slashing             |
| Stake lock period         | None         | Can unregister and withdraw anytime      |

### Why Stake?

- **Sybil resistance:** Prevents spam registrations
- **Quality signal:** Hosts with higher stakes demonstrate commitment
- **Weighted selection:** SDK's host selection algorithm weights stake at 35% by default
- **Dispute collateral:** Stakes can be slashed for proven misbehaviour (evidence-based)

### Slashing at MVP:

The protocol includes owner-controlled stake slashing at launch. When misbehaviour is proven via the CID evidence trail:

1. Owner reviews evidence (proofCID, deltaCID, conversationCID)
2. Owner calls `slashStake(host, amount, evidenceCID, reason)`
3. Slashed tokens transfer to protocol treasury
4. All slashes emit public events for transparency

See [SLASHING\\_SPECIFICATION.md](#) for full implementation details.

## 7.5 Host Selection and Reputation

The SDK implements a **weighted host selection algorithm** that users can configure:

| Factor         | Default Weight | Source                                               |
|----------------|----------------|------------------------------------------------------|
| <b>Stake</b>   | 35%            | On-chain <code>stakedAmount</code> from NodeRegistry |
| <b>Price</b>   | 30%            | Normalised inverse of host pricing                   |
| <b>Uptime</b>  | 20%            | Placeholder (95%) until metrics system               |
| <b>Latency</b> | 15%            | Placeholder (100ms) until metrics system             |

### Selection Modes:

- **AUTO:** Standard weighted scoring (default)
- **CHEAPEST:** 70% price weight
- **RELIABLE:** 50% stake + 40% uptime weight
- **FASTEST:** 60% latency weight
- **SPECIFIC:** Use a preferred host address

Note: Uptime and latency currently use placeholder values. A comprehensive metrics collection system is planned for future development.

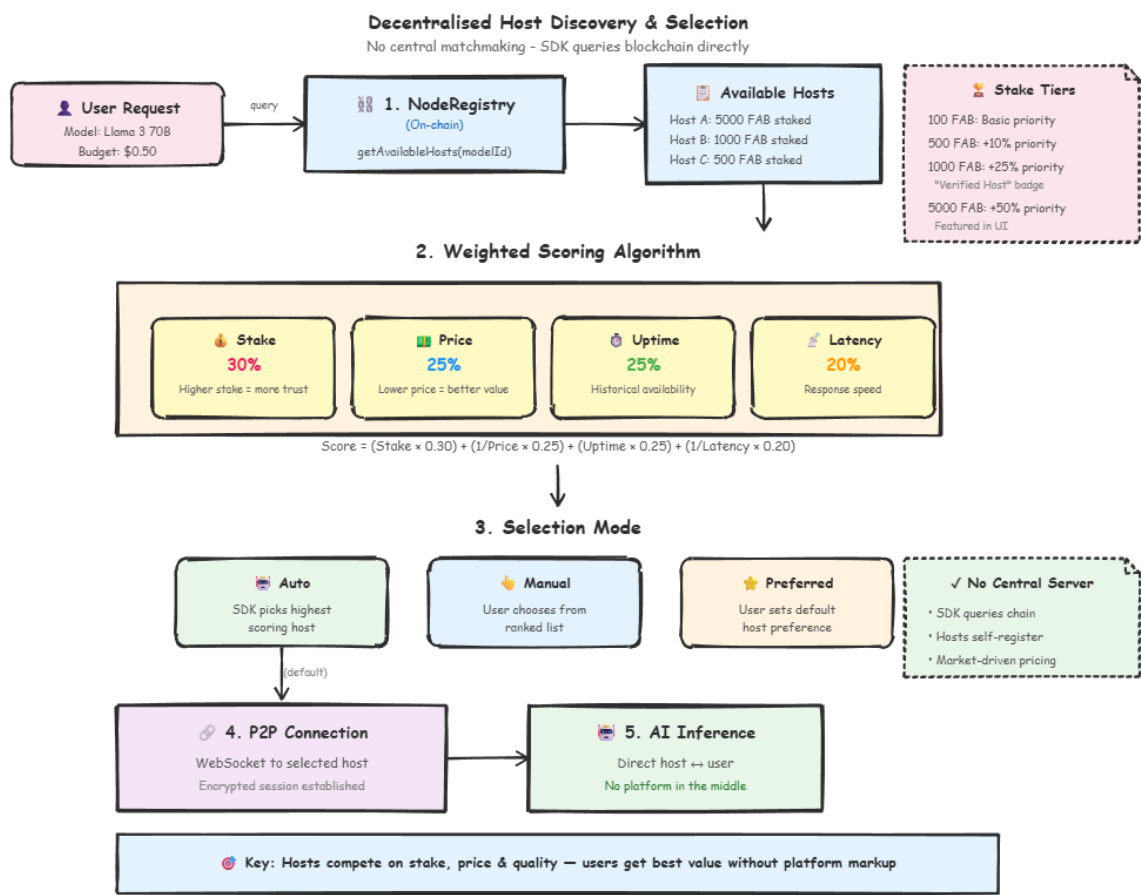


Figure 6: Host Selection Algorithm showing the decentralised discovery and selection flow from NodeRegistry query through weighted scoring (stake, price, uptime, latency) to P2P connection.

7.6 Minimum Deposits

| Payment Method | Minimum              | Rationale             |
|----------------|----------------------|-----------------------|
| ETH            | 0.0001 ETH (~\$0.40) | Cover gas costs       |
| USDC           | \$0.50               | Prevent dust sessions |

These minimums ensure sessions are economically viable for hosts.

7.7 Delegated Payments (Sponsored Sessions)

For agentic and SaaS use cases, a third party often needs to pay for inference on behalf of its users or autonomous agents — a SaaS backend covering its customers' usage, or a treasury sponsoring a fleet of coding agents — without surrendering spending control. The protocol supports this through **delegated payments** (delegate-pays):

- A **payer** account pre-approves a bounded USDC allowance to the **JobMarketplace** contract and authorises one or more **delegate** addresses on-chain (**authorizeDelegate**).
- A **delegate** — typically an ephemeral hot key generated for a bridge or daemon — signs sessions and spends *from the payer's allowance* but **never holds the funds**. The on-chain

allowance is a hard ceiling: a runaway agent cannot spend beyond it.

- The SDK enters this mode via `authenticateAsDelegate({ signer, payer })`; sessions are created through `createSessionForModelAsDelegate`, with pre-flight allowance and balance checks before any chain write. Delegated sessions are **USDC-only**.
- Because an ERC-20 allowance is shared across a payer's delegates, the payer should be an account distinct from any interactive wallet. The orchestrator daemon wires this up from a single `FABSTIR_PAYER` environment variable, exposing an OpenAI-compatible endpoint that agents call while the payer foots the metered, capped bill.

This makes Platformless AI usable as a sponsored backend: end users and autonomous agents consume decentralised inference without managing wallets or gas, while the sponsor retains a cryptographically enforced spending cap. It composes with the x402 inter-agent payment layer (Section 6.5) — delegate-pays funds a party's own hosted sessions, while x402 settles cross-agent delegation.

## 8. Security and Privacy

### 8.1 Threat Model

The protocol defends against:

| Threat                           | Mitigation                                                                                                                                                                                          |
|----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Eavesdropping                    | End-to-end encryption                                                                                                                                                                               |
| Man-in-the-middle                | ECDSA authentication                                                                                                                                                                                |
| Replay attacks                   | Nonces, timestamps, message indices                                                                                                                                                                 |
| Malicious hosts                  | Staking + evidence-based slashing + cryptographic proofs                                                                                                                                            |
| Data theft                       | User-controlled encryption keys                                                                                                                                                                     |
| Host reads inputs/weights in-use | Confidential computing / TEE: attested key release, model decrypted only inside the confidential VM (proven on real TDX + H200 hardware with paid testnet sessions, not yet the default path, §8.6) |
| Platform capture                 | Decentralised storage + blockchain                                                                                                                                                                  |
| Censorship                       | Permissionless participation                                                                                                                                                                        |
| Illegal content distribution     | Fail-closed hash-matching gates on the media pipeline + evidence quarantine + human review (§8.7)                                                                                                   |

### 8.2 Encryption Architecture

All sessions are encrypted by default using state-of-the-art cryptography.

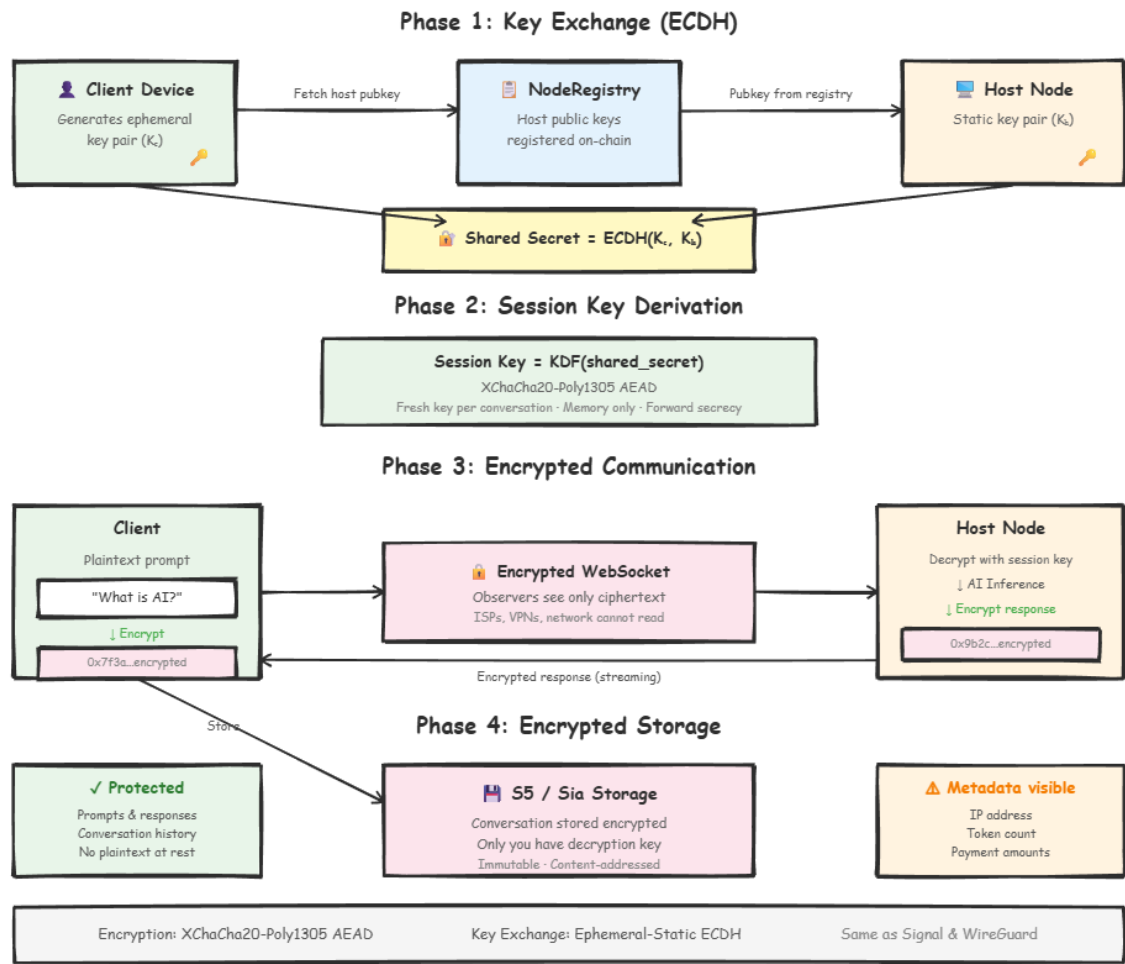
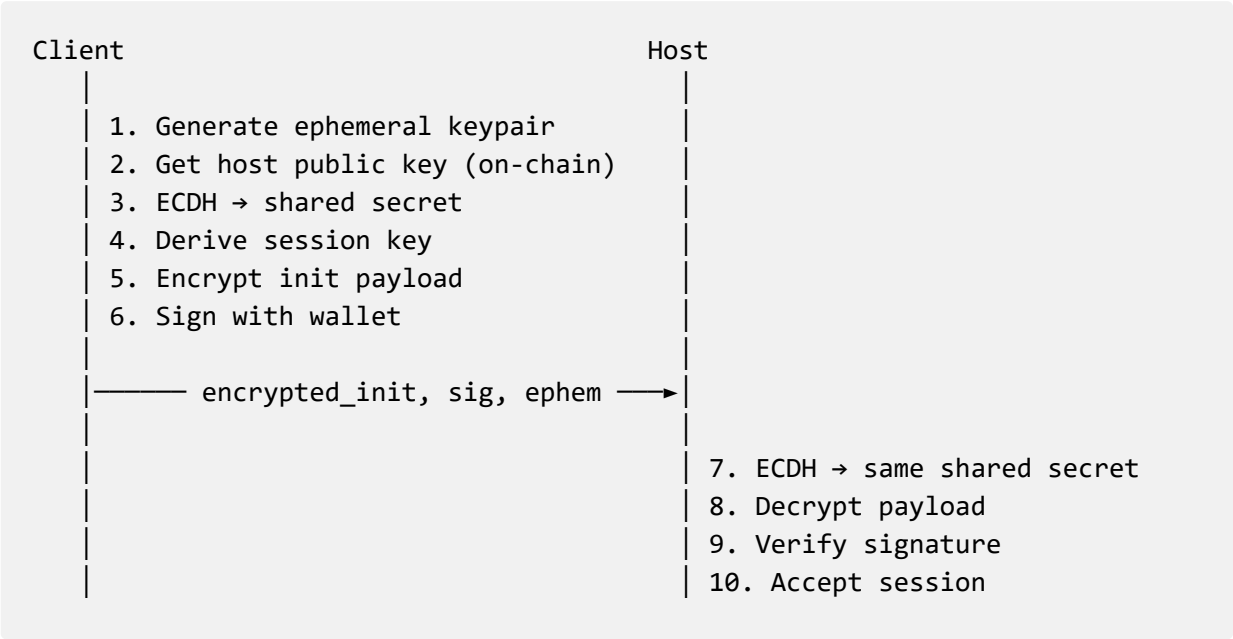


Figure 7: End-to-End Encryption Flow showing ECDH key exchange, session key derivation, and encrypted communication.

Session Initialisation (Ephemeral-Static ECDH):



Message Encryption (Symmetric):

After session init, messages use fast symmetric encryption:

```
encrypt(session_key, message, index) → ciphertext
```

- **XChaCha20-Poly1305**: AEAD cipher with 256-bit keys
- **Message index in AAD**: Prevents replay
- **~1ms overhead**: Negligible performance impact

#### Forward Secrecy:

Ephemeral keys are generated fresh per session and discarded after use. Even if host's long-term key is compromised, past sessions remain secure.

## 8.3 What's Protected

#### ✓ Encrypted:

- Prompts and responses
- Conversation history (on S5)
- Session configuration
- RAG queries and context

#### ✗ Visible on-chain:

- Session creation/completion
- Payment amounts
- Token usage
- Host selection

On-chain data is necessarily public for verification, but contains no content.

### 8.3.1 In-Use Confidentiality (Current Boundary)

End-to-end encryption protects content **in transit** (client ↔ host) and **at rest** (on S5). It does not, by itself, make computation blind: the host the user selects necessarily decrypts the prompt in memory to run inference. "End-to-end" here means client to host with no platform intermediary in between — not that the host computes on ciphertext.

What this means in practice:

- Plaintext exists only **transiently, in the chosen host's memory**, for the duration of inference, and is never persisted unencrypted. No platform, proxy, or storage node ever sees it.
- Hosts are stake-bonded, independent operators selected by the user; a stateless, non-logging host that retains nothing is a far weaker exposure than a centralised provider that logs every query indefinitely.
- For workloads that require the host to be blind to inputs as well, two paths apply:  
**confidential computing via Trusted Execution Environments (TEE)**, the route to in-use privacy, now proven end to end on real confidential-computing hardware (Intel TDX with an NVIDIA H200), with paid testnet sessions run on it, but not yet the default path (Section 8.6); and, today, **client-side pre-processing** — sensitive inputs can be embedded or transformed locally so only derived data (e.g. vectors) reaches the host.

**Fine-tuning widens this boundary, and the difference is worth stating rather than leaving implied.** A training job hands the host the user's dataset, not a single prompt, and the host must decrypt it to tokenise and train on it. It is also on disk rather than only in memory: shards are written to a staging volume for the duration of the run. The protocol narrows the window rather than closing it — each shard is encrypted under its own key in transit and at rest on S5, staged files are removed on every terminal path and swept at start-up, and the adapter is re-encrypted before it leaves — but for the length of a run the host holds the user's corpus in clear. The honest summary is that training today has the same trust boundary as inference, applied to a larger and more sensitive input, and that TEE is the route which closes it rather than a mitigation already in place. That route has now been exercised: on 30 September 2026 a training job ran on the attested confidential-VM host of Section 8.6, where the dataset was decrypted only inside the confidential VM. It is not yet the default path, and the open question in Section 8.6 about GPU memory applies to training exactly as it does to inference.

We state this boundary explicitly because honest scoping of cryptographic guarantees is itself a security property.

## 8.4 Key Management

### Client Keys:

Derived from wallet private key using deterministic signature-based derivation. No additional secrets to manage.

### Host Keys:

Registered on-chain during node setup. Public key stored in NodeRegistry metadata.

### Session Keys:

Generated fresh per session, exist only in memory, discarded on completion.

## 8.5 S5 Storage Encryption

Data stored on S5 is encrypted by the SDK before upload:

```
// SDK encrypts before storage
const encrypted = encryptionManager.encrypt(conversation);
await storageManager.save(encrypted);

// S5 stores opaque bytes—cannot read content
```

S5 nodes see only encrypted blobs. Only users with the correct keys can decrypt.

## 8.6 Confidential Computing and Trusted Execution Environments (Proven on Hardware, Pre-Production)

Section 8.3.1 stated the current boundary plainly: the host a user selects must decrypt the prompt in memory to run inference, so end-to-end encryption protects content **in transit** and **at rest** but

not **in use**. **Confidential computing** is the protocol's route to closing that last gap — *host-blind inference*, where even the host's own root operator cannot read the prompt or the model weights.

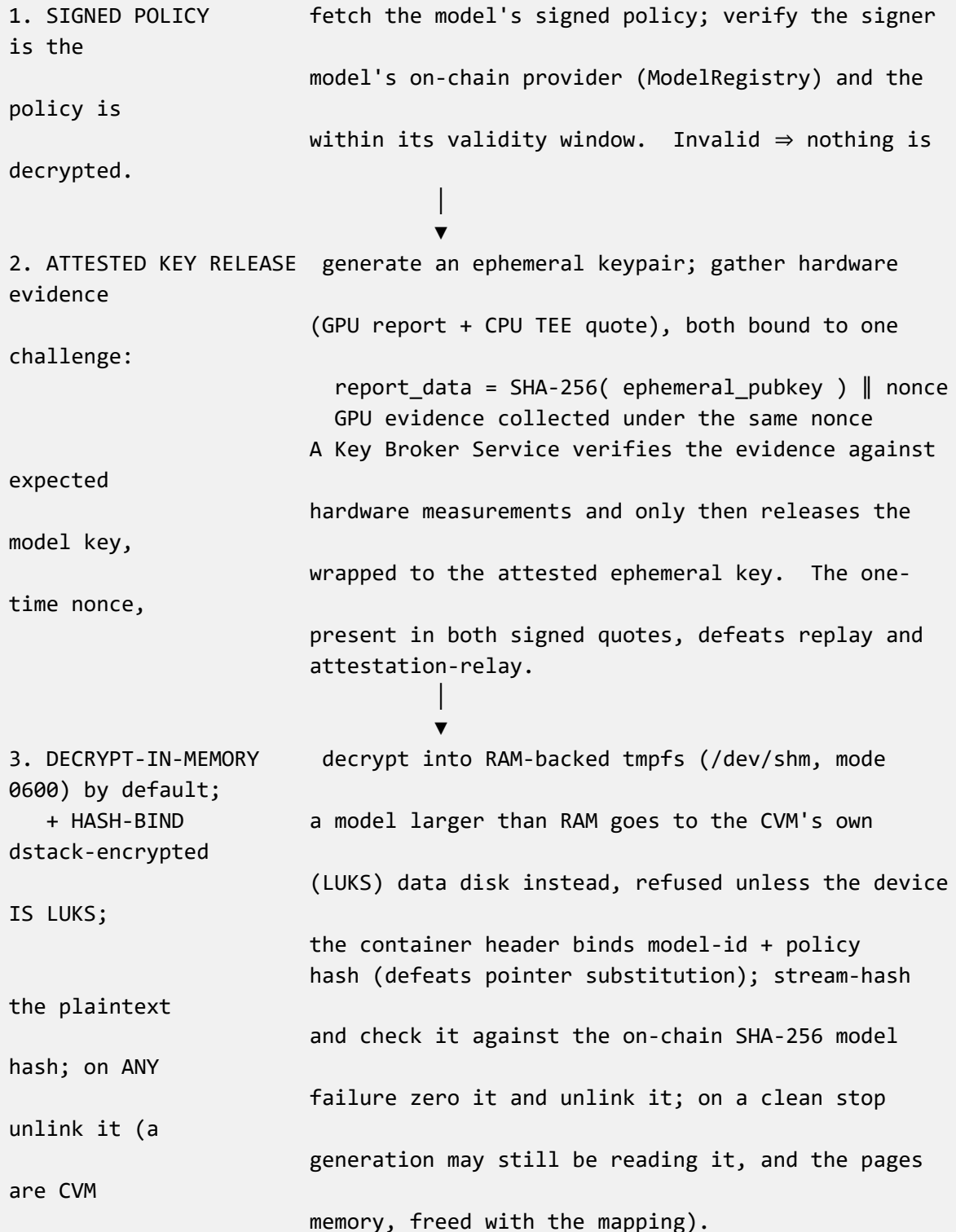
The implementation runs in the node binary's live request path, and on 23 September 2026 it ran end to end on real confidential-computing hardware: a Phala Cloud confidential VM on Intel TDX with an NVIDIA H200 GPU in confidential-computing mode. An off-node Key Broker Service verified the TDX quote and the H200's attestation through NVIDIA's Remote Attestation Service, both carrying the same one-time challenge, against a provider-signed policy, and only then released the model key. The model, the open-weight Qwen3.8-27B (29 GB), was decrypted inside the confidential VM, loaded onto the H200 and served inference. The same day, paid testnet chat sessions from the Platformless AI app were routed to that host, checkpointed and settled on-chain, with their proofs published to decentralised storage. We describe it here because the architecture is built and proven on the target hardware, and the trust model it enables is central to the protocol's privacy story, while being explicit about what it does not yet prove (see *Honest status* below) and that it is not yet the default path for paid sessions. A week later, on 30 September 2026, a second run on the same hardware put the whole product on one such machine: chat, image generation, fine-tuning with the adapter served back, and video generation from the Blender extension, every service deployed in one confidential VM whose language-model key was released through the same attested path.

### What it adds beyond end-to-end encryption

With confidential computing, the model is served from a hardware-isolated **Trusted Execution Environment (TEE)**: a Confidential VM combining a **CPU TEE** (Intel TDX or AMD SEV-SNP, which encrypts and integrity-protects the VM's memory) with an **NVIDIA GPU in Confidential Computing mode**, which NVIDIA designs to protect the PCIe link and GPU memory from the host. What the protocol can state today, and what the hardware run demonstrated, is narrower and exact: the model key is released only against hardware-signed proof that the machine is genuine and running approved software; the model is decrypted only inside a confidential VM whose memory the operator cannot read; and the GPU proves its identity, secure boot and debug state to NVIDIA's verification service. The further claim, that the operator cannot read GPU memory while the model runs, depends on the GPU being in confidential-computing mode, which the signed evidence does not yet settle (see *Honest status*). Trust shifts from "*trust the host operator not to look*" towards "*trust the silicon vendor's hardware root of trust, verified by a remote attestation that the client can check*".

### The attested-model pipeline

Models are distributed **encrypted** (a sealed AES-GCM container whose data-encryption key the host never holds). Serving such a model runs through three sequential, **fail-closed** gates (**tee::prepare\_attested\_model**):



Every step fails closed: the model key is released **only** after a valid attestation; a node that is not TEE-enabled **refuses encrypted models outright**; and a host advertises the **tee-attested** capability — in its on-chain **NodeRegistry** registration and in the encrypted-WebSocket handshake — **only** when TEE enforcement is actually active. The same single flag gates both advertisement and enforcement, so a host cannot claim attested confidentiality it does not honour. Clients can therefore filter host discovery for **tee-attested** nodes and route sensitive workloads only to hosts whose model key was released against hardware-signed proof of a genuine confidential VM running approved software. This composes with the existing layers — E2E encryption still protects transit and storage; model governance (Section 4.2) still pins the model's hash and provider on-chain; the TEE adds in-use confidentiality on top.

## Honest status

- **Pipeline built.** The pieces — sealed-container format, signed-policy verification, ephemeral-key remote attestation with cross-binding, Key-Broker key release, tmpfs-only decryption, on-chain hash-binding, secure delete, and capability advertisement — are implemented, fail-closed throughout, and covered by more than 300 passing tests across the node and key-broker suites.
- **Proven on real GPU hardware, end-to-end in integration.** The complete encrypted-model path runs green on an NVIDIA GPU host from an integration test: encrypt → attest → key release → decrypt to `/dev/shm` → `llama.cpp` inference on CUDA → secure delete.
- **The CPU half now runs on real silicon.** On a confidential VM with an Intel TDX processor, the node produced a genuine TDX quote, an off-node Key Broker Service verified it against a provider-signed policy and released the wrapped model key in well under a second, and the node decrypted the container and served an inference. The launch measurement and the boot-time registers are pinned; the runtime register is replayed from the VM's event log rather than trusted; the quote's certification data is checked against Intel-signed collateral; and the one-time nonce is required in both halves of the evidence. This is the live request path in the shipped binary, not a simulator and not an integration harness, and the key-broker, policy-source and blob-source seams are now real implementations rather than mocks.
- **The GPU half now runs on real silicon (23 September 2026).** On a Phala Cloud confidential VM pairing Intel TDX with an NVIDIA H200, the key broker verified a genuine TDX quote (Intel TCB status up to date, no advisories) and the H200's own attestation report through NVIDIA's Remote Attestation Service (hardware model, driver and VBIOS versions, secure boot on, debug disabled), and the same one-time challenge was confirmed in both the CPU quote and NVIDIA's signed result. It released the model key on the first boot after the machine's measurements were pinned. The confidential VM's launch and boot measurements were recomputed independently from the provider's published image and matched the hardware quote before they were pinned, rather than simply trusted on first use. The node then checked the decrypted model against its on-chain hash, loaded all 66 layers onto the H200 and served coherent inference through the provider's gateway; on a clean stop it deleted the plaintext. A replayed key request was refused. The broker held the real key ring, not the throwaway ring used in the CPU rounds.
- **What the GPU evidence does and does not settle.** NVIDIA publishes no single field saying "confidential computing is on". The signed per-GPU claims for secure boot and debug state together exclude the developer-tools mode in which the host can inspect GPU memory, and the key broker enforces that pair as a condition of release; the H200 run confirmed both values on real hardware. They do not distinguish confidential computing being switched on from being switched off, and inspecting the H200's signed report on the day found no field that does. Whether a Hopper GPU can produce evidence that NVIDIA's service will pass with confidential computing switched off is not documented either way, and the question is with NVIDIA. Until it is answered, that distinction rests on the platform provider's configuration plus a check inside the measured image, which read confidential computing as on throughout the run. **So this work still does not prove that a root operator cannot read GPU memory**, and that claim is deliberately absent from the rest of this document. This is recorded as an open gap rather than papered over.

- **Paid sessions from the app, on the same hardware.** The same day, with the host's on-chain registration pointed at the confidential VM by hand, the Platformless AI app ran three encrypted testnet sessions on it; for each, the node generated a real zero-knowledge proof, submitted the checkpoint and completed the session on-chain, all confirmed. In a further session with the storage bridge added to the confidential VM, the proof was published to the S5 network: fetched there by its content identifier from an unrelated machine, its SHA-256 equals the proof hash the host committed on-chain.
- **The whole product on one attested machine (30 September 2026).** A second H200 run deployed the inference node, image generation (FLUX.2), the fine-tuning trainer, the storage bridge and the video pipeline (LTX 2.3) as one configuration. Its configuration hash was computed in advance from our own compose file and signed into the policy before the machine existed; it matched on the first boot, as did the launch registers pinned after the first run, and the key was released. Changing the configuration later that day (a storage-bridge fix and the video services) changed the hash, so release required a newly signed policy naming the new hash and a container resealed under it; the encrypted copy cached on the machine, bound to the previous policy, was detected as stale and replaced. Over three boots the key was released three times, each against a fresh quote and fresh GPU evidence. From the app and the Blender extension: encrypted chat, images, a fine-tune whose dataset was decrypted only inside the confidential VM and whose adapter came back encrypted and was then served into a session on the same machine, and six video-generation modes; every job settled on-chain. Two independent fine-tunes of the same job produced byte-identical adapters.
- **Remaining milestones.** Answer the on-versus-off question above; route paid marketplace sessions to **tee-attested** hosts automatically rather than by hand; bring the video-generation (LTX) weights under the same attested release, since today only the language model's weights are sealed and bound (on 30 September the video weights were verified against pinned hashes as they downloaded into the confidential VM, but they are public weights that the broker neither seals nor releases); separate the approved host measurements from the sealed model binding, so that approving a new host type no longer forces every model to be resealed; add an independent second signer for those host approvals, which first needs the node image to be reproducible from public source; and cross-check the pinned measurements against the provider's own published values. **Until then, confidential inference should be treated as proven on the target hardware but not yet a production guarantee.**

This is the strongest privacy posture the protocol targets, and the natural fit for regulated and IP-sensitive workloads — healthcare, legal, and enterprise source code — where even a trusted, stake-bonded host must be unable to read the inputs or the proprietary model weights it serves.

---

## 8.7 Content Moderation

Decentralisation does not remove the obligation to keep known illegal content off the network, and the protocol's answer is designed to preserve its other properties while meeting that obligation.

The operating layer is **deterministic hash matching** on the media pipeline: during every video transcode, sampled keyframes are matched against block lists of confirmed illegal content, using exact matching (SHA-256) on source files and perceptual matching (PDQ) on frames, which survives

re-encoding. This involves no AI judgment of the content — a verdict is a reproducible fact about a file against a list, recorded with a fingerprint of the exact list that produced it.

Three independent, fail-closed gates act on verdicts: the host node withholds completion and payment for non-cleared jobs; the transcoding service withholds upload of playable output; and the client SDK refuses the publish step that makes content visible. Each gate fails closed on its own inputs — an unreachable scanner or unloadable list results in a hold, never a silent pass. An absent verdict always means "not scanned", never "clean".

Matched material is preserved in an encrypted evidence quarantine and is never auto-reported: a human reviewer examines each case, including which list produced the match, before anything is filed with the applicable reporting body. Moderation reads only what the processing host can already see during the work it was hired to do — it creates no new access to user content, and verdicts are local to the host that produced them; the protocol imposes no central moderation authority on the network.

A second, complementary layer — retrospective AI-based scanning of novel content — is built but deliberately not enforcing, pending a policy design that does not misclassify lawful material. Enforcement of both layers is staged: observe first, enforce after operational soak.

## 9. Dispute Resolution

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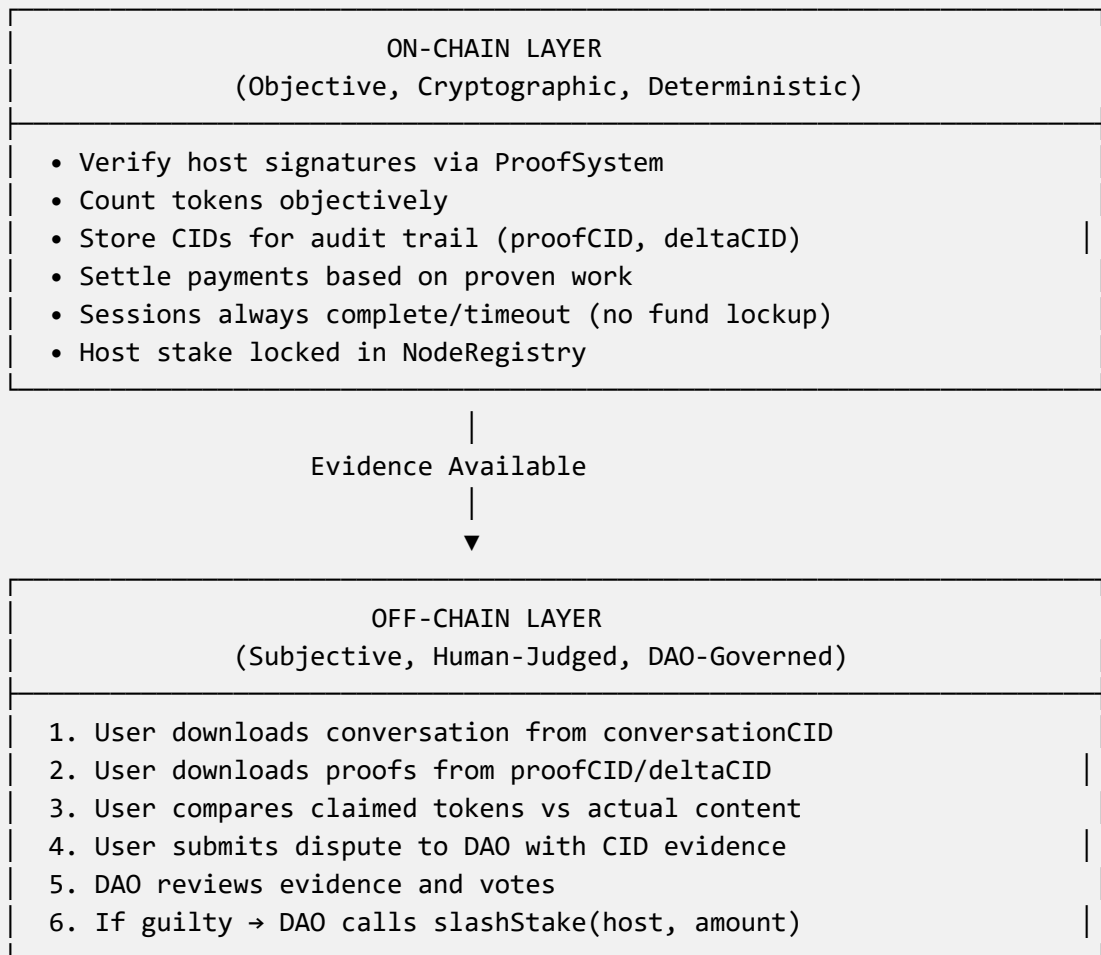
### 9.1 Design Philosophy

The protocol takes a **pragmatic approach** to disputes:

- **On-chain:** Handles objective, cryptographically verifiable facts
- **Off-chain:** Handles subjective quality judgements (future DAO)

Smart contracts **cannot** determine if an AI response was "good quality" or "factually correct"—these are inherently subjective. Attempting on-chain dispute resolution creates complexity, attack vectors, and gas waste.

### 9.2 Two-Layer Architecture



### 9.3 Evidence Trail

Every proof submission creates an immutable, verifiable record:

| Component                    | Storage  | Purpose                         |
|------------------------------|----------|---------------------------------|
| <code>proofHash</code>       | On-chain | Cryptographic fingerprint       |
| <code>proofCID</code>        | S5       | Complete proof data             |
| <code>deltaCID</code>        | S5       | Incremental changes             |
| <code>conversationCID</code> | S5       | Full conversation at completion |
| <code>signature</code>       | On-chain | Host's ECDSA attestation        |

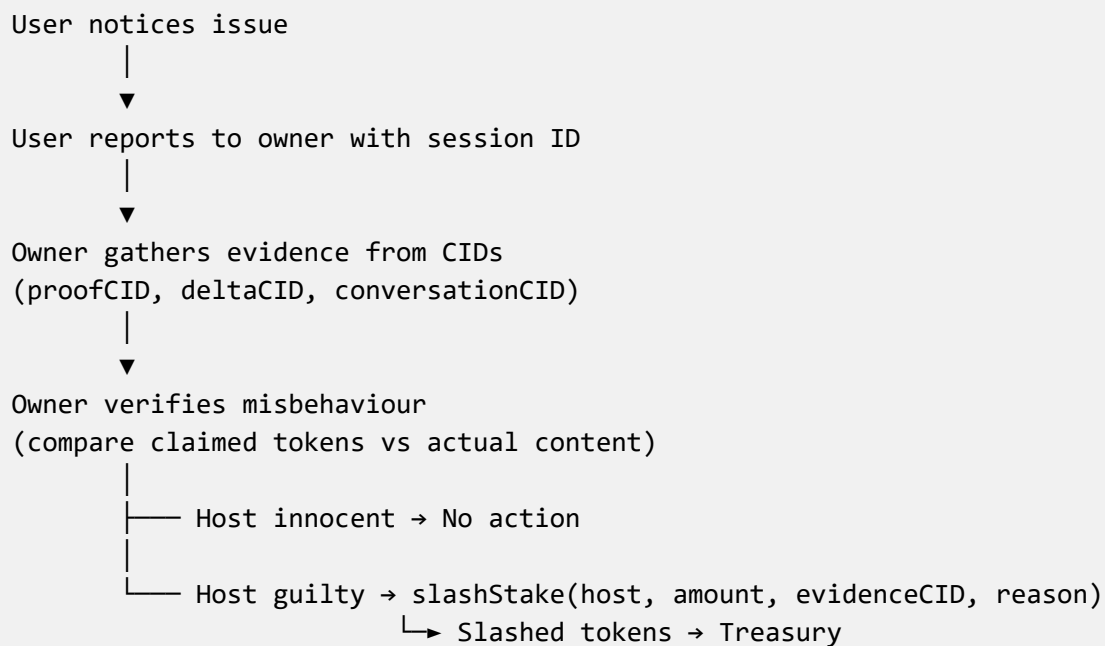
#### Anyone can verify:

1. Download CID content from S5
2. Hash content, compare to on-chain `proofHash`
3. Verify signature matches host
4. Count actual tokens vs claimed tokens

### 9.4 Dispute Resolution Flow

#### MVP: Owner-Controlled Slashing

At launch, dispute resolution is handled by the protocol owner using evidence-based slashing:



### Safety Constraints:

- Maximum 50% slash per action
- 24-hour cooldown between slashes on same host
- Evidence CID required (public accountability)
- All slashes emit events for transparency

### Future: DAO-Governed Slashing

Post-MVP, slashing authority transfers to a DAO contract:

```
// Owner transfers authority to DAO
setSlashingAuthority(daoContractAddress);
```

The DAO then manages dispute submissions, voting, and slash execution. No contract upgrade required—same `slashStake()` function, different caller.

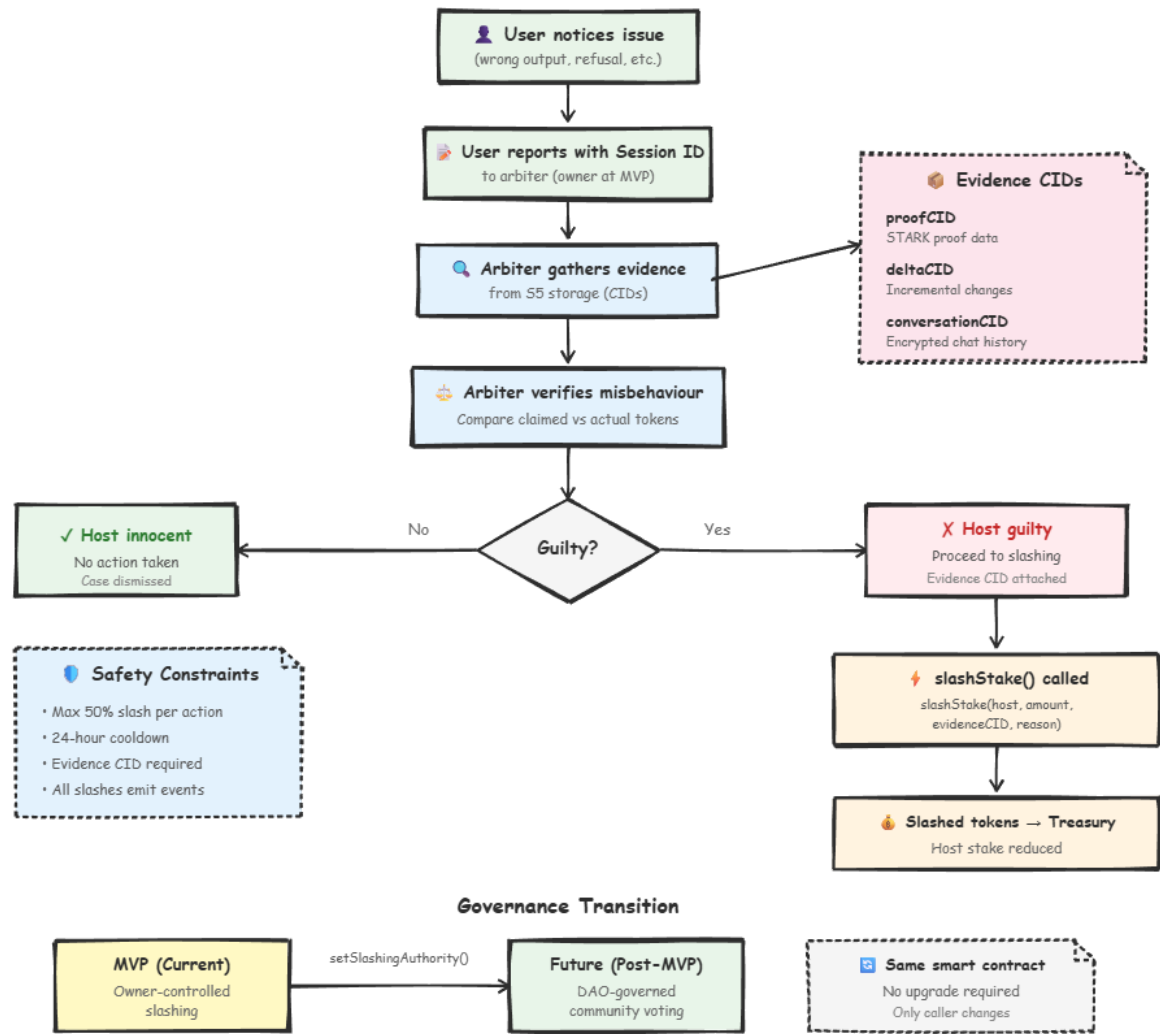


Figure 8: Dispute Resolution Flow showing the evidence-based slashing process from user report through arbiter verification to outcome, with safety constraints and governance transition path.

## 9.5 Attack Mitigation

| Attack                       | Mitigation                                          |
|------------------------------|-----------------------------------------------------|
| Host misbehaviour            | Evidence-based slashing with CID proof              |
| False evidence               | CIDs are content-addressed (immutable, verifiable)  |
| Fund lockup grieving         | Sessions always settle; no on-chain dispute state   |
| Slash abuse                  | 50% max per action, 24h cooldown, evidence required |
| Arbiter centralisation (MVP) | All slashes public, DAO transition planned          |

## 10. Governance

### 10.1 Current State

At MVP launch, governance operates in a **bootstrap phase** with admin-controlled operations:

- Contract upgrades controlled by admin keys (single-sig initially)
- Model approval via owner functions (`addTrustedModel`)
- Stake slashing via owner functions (`slashStake`)
- Parameter changes via owner functions

This centralisation is temporary and necessary for rapid iteration during early deployment. All admin actions are on-chain and publicly verifiable.

## 10.2 Progressive Decentralisation

The roadmap includes progressive decentralisation:

| Phase    | Governance                               |
|----------|------------------------------------------|
| MVP      | Admin-controlled (bootstrap)             |
| Post-MVP | Multi-sig with community representatives |
| Mature   | Full FAB token-weighted DAO voting       |

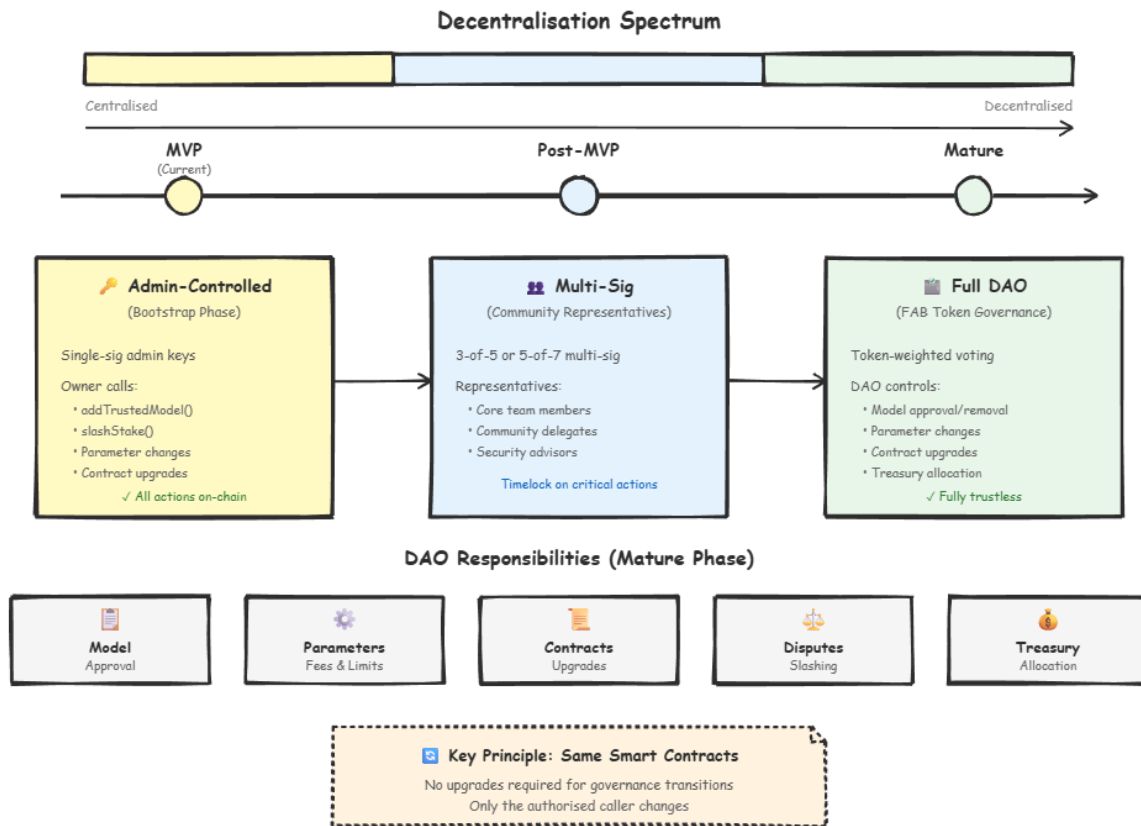


Figure 9: Progressive Decentralisation Timeline showing the governance transition from admin-controlled MVP through multi-sig to full DAO governance, with key responsibilities at each phase.

## 10.3 DAO Responsibilities (Future)

- Model approval/removal via voting
- Parameter changes (fees, minimums)
- Contract upgrades
- Dispute resolution and stake slashing

- Treasury allocation

## 10.4 Model Governance

The ModelRegistry contract is **architecturally ready** for community governance with two approval tiers:

### Tier 1: Owner-Curated (MVP)

```
// Owner adds trusted models directly
function addTrustedModel(
    string huggingfaceRepo,
    string fileName,
    bytes32 sha256Hash
) external onlyOwner;
```

### Tier 2: Community-Voted (Available, No UI Yet)

```
// Anyone can propose a model (requires FAB stake)
function proposeModel(string repo, string file, bytes32 hash) external;

// FAB token holders vote
function voteOnProposal(bytes32 modelId, uint256 amount, bool support)
external;

// Execute passed proposals
function executeProposal(bytes32 modelId) external;
```

The community voting infrastructure exists in the deployed contracts but awaits UI implementation. For MVP, the owner uses `addTrustedModel()` to bootstrap the model ecosystem.

### Model Governance Prevents:

- Malicious models (trojaned weights)
- Illegal content generation models
- Models violating licensing terms

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## 11. Roadmap

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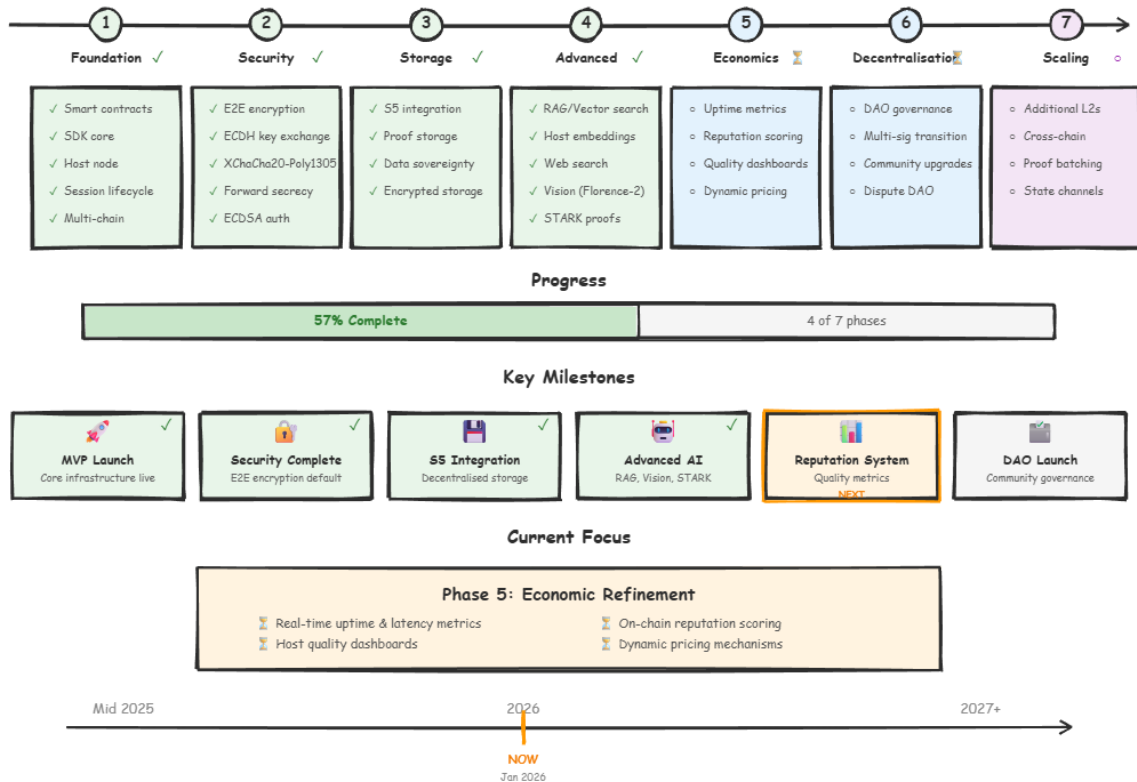


Figure 10: Roadmap Timeline showing development progress across 9 phases, with 6 phases completed (67%) and current focus on Phase 7: Economic Refinement.

## Phase 1: Foundation (Completed)

- ✓ Smart contract deployment (UUPS upgradeable)
- ✓ SDK core functionality
- ✓ Host node implementation
- ✓ Basic session lifecycle
- ✓ Multi-chain support (Base Sepolia)

## Phase 2: Security (Completed)

- ✓ End-to-end encryption (default)
- ✓ Ephemeral-static ECDH key exchange
- ✓ XChaCha20-Poly1305 symmetric encryption
- ✓ Forward secrecy
- ✓ ECDSA authentication

## Phase 3: Storage (Completed)

- ✓ S5 integration for conversations
- ✓ Proof storage (proofCID, deltaCID)
- ✓ User-owned data sovereignty
- ✓ Encrypted storage

## Phase 4: Advanced Features (Completed)

- ☒ RAG/Vector search support
- ☒ Host-side embeddings
- ☒ Web search integration
- ☒ Vision processing (Florence-2, OCR)
- ☒ Image generation (FLUX.2 diffusion sidecar with SDK auto-intent detection)
- ☒ AI video generation (LTX 2.3 via ComfyUI sidecar) — **thirteen modes**: text-to-video, image-to-video, first-last-frame, ingredients, restyle, outpaint, edit, restore, upscale, water, day-to-night, cross-view, and SDR-to-HDR (the guided ones on the IC-LoRA union-control path); pinned-template provenance with byte-exact input binding for stills and control video (versioned per input kind); megapixel-frame billing; **full economics live**: host-side on-chain proof submission, dispute-window settlement, 90/10 split — paid sessions settled on Base Sepolia and verified to the unit from the SDK, the product UI, and Blender; **4K live-proven** (a 5-second 3840×2160 clip settled at 1.0M billed tokens in 8m15s), with a per-job token cap rather than a resolution cap; allow-list bundle v25 carries the /64-exact rungs the model can actually deliver; encrypted input stills to 32 MB and control clips to 128 MB; **all thirteen modes live inside Blender** — a native 5.x extension in the Video Sequence Editor, clips generated and settled on-chain to the unit from within the timeline, with conform-rendered control clips, frame-aligned audio, and optional archive of delivered clips to the user's own encrypted Sia account — next: signed attestations, C2PA export
- ☒ Zero-knowledge proof generation (host-side STARK proofs)
- ☒ GPU-accelerated video transcoding (H.264, HEVC, AV1 via NVENC sidecar)
- ☒ End-to-end encrypted video pipeline (source/output encrypted on Sia, E2EE WebSocket transit)
- ☒ Capacity-aware transcode load balancing across multiple hosts
- ☒ Per-transcode WebSocket isolation for concurrent job support
- ☒ Pending job tracking with automatic overflow and retry-with-backoff
- ☒ GOP-level verification layer (STARK proofs, Merkle tree, on-chain root commitment)
- ☒ Stress-tested: 7 concurrent jobs across 2 GPU hosts (NVIDIA L40S + RTX Pro 6000), 100% completion
- ☒ HLS adaptive bitrate streaming — fMP4 segments with per-segment encryption (SDK v1.18.0)
- ☒ Client-side M3U8 playlist generation (master + variant, HLS v7 with fMP4 init segment)
- ☒ Per-segment encryption: preview segments unencrypted (z-prefix), paid segments XChaCha20-Poly1305 encrypted (u-prefix)
- ☒ End-to-end validated: encrypted upload → GPU transcode (NVIDIA L40S) → HLS segments on Sia/S5 → USDC payment → hls.js playback with decrypting service worker

*Note: Production UI for ZK proof verification display is pending; backend infrastructure is complete.*

## Phase 5: Agentic AI Infrastructure (Completed)

- ☒ Claude Bridge — Anthropic Messages API compatibility layer
- ☒ OpenAI Bridge — OpenAI Chat Completions, Images, and Responses API compatibility layer
- ☒ Full tool use support (23+ tools, streaming SSE, multi-turn tool results)
- ☒ Image generation via [/v1/images/generations](#) (FLUX.2 diffusion, quality/size mapping)
- ☒ Vision support (base64 image\_url passthrough)

- ✓ Think-block stripping and output safeguards
- ✓ Session auto-recovery on disconnect
- ✓ Production validation: Claude Code and OpenCode running on decentralised hosts
- ✓ SaaS-ready: drop-in Anthropic or OpenAI API replacement for product teams
- ✓ 129 + 162 tests passing

## Phase 6: Multi-Agent Orchestration (Experimental — Completed)

- ✓ @fabstir/orchestrator package — DAG-based task decomposition and execution
- ✓ A2A protocol (v1.0.0-rc) — Agent Cards, skill-based discovery, JSON-RPC delegation
- ✓ Intelligent model routing — task-type-aware assignment with on-chain validation
- ✓ Orchestration patterns — FanOut, Pipeline, MapReduce
- ✓ SSE streaming — real-time phased progress, content negotiation, task cancellation
- ✓ x402 HTTP payment — USDC micropayments for inter-agent delegation (EIP-3009)
- ✓ Session pooling — semaphore-based concurrency with nonce serialisation
- ✓ Session multiplexing — per-model session caching to reduce USDC deposit burden (up to 67% savings)
- ✓ BigInt consistency — uint256 ABI params consistently encoded as BigInt for Account Kit compatibility
- ✓ 351 tests passing

## Confidential Computing — Trusted Execution Environments (Proven on Hardware, Pre-Production)

*A parallel security workstream toward host-blind inference (Section 8.6). Proven end to end on real confidential-computing hardware (Intel TDX with an NVIDIA H200, September 2026); paid testnet sessions from the app have run on it; on 30 September 2026 the whole product (chat, images, fine-tuning with serve-back, and video from Blender) was served from one such machine; automatic routing, sealed video weights and one open GPU-evidence question remain.*

- ✓ Encrypted-model container format (sealed AES-GCM; the host never holds the key)
- ✓ Signed model policy — provider-signed, on-chain-provider verification, validity window
- ✓ Remote-attestation key release — ephemeral keypair + GPU report + CPU-TEE quote, both bound to one challenge (`report_data = SHA-256(pubkey) || nonce`; GPU evidence collected under the same nonce), one-time nonce; Key Broker releases the model key only on a valid attestation
- ✓ Decryption to RAM-backed tmpfs (`/dev/shm`) by default, or for a model larger than RAM to the CVM's dstack-encrypted (LUKS) data disk, refused unless the device is LUKS; the plaintext is zeroed and unlinked on any failure, and unlinked on a clean stop
- ✓ On-chain SHA-256 model hash-binding; container-header model-id/policy binding (anti-substitution)
- ✓ Fail-closed throughout — non-TEE nodes refuse encrypted models; the `tee-attested` capability gates advertisement and enforcement with one flag
- ✓ Proven end-to-end in integration on real NVIDIA GPU hardware (encrypt → attest → key release → decrypt → `llama.cpp` on CUDA → inference → secure delete); 300+ tests; adversarial review rounds throughout

-  **Real Intel TDX attestation, live:** a genuine hardware quote from a confidential VM, verified by a deployed off-node key-broker service against a provider-signed policy, releasing the model key to the node's live request path
-  **Real NVIDIA H200 attestation, live (23 September 2026):** on a Phala Cloud confidential VM, the key broker verified the TDX quote and the H200's report through NVIDIA's Remote Attestation Service under one challenge, released the key on the first pinned boot, and the open-weight Qwen3.8-27B was decrypted inside the confidential VM, loaded onto the H200 and served inference; measurements recomputed independently before pinning; replay refused
-  **Open question with NVIDIA:** whether any signed evidence distinguishes confidential-computing mode on from off on Hopper. Until it is answered, this document does not claim that a root operator cannot read the model or prompt from GPU memory
-  Paid testnet sessions from the Platformless AI app run on the TEE host end to end (encrypted chat, zero-knowledge proof, on-chain checkpoint and settlement), with the host routed by hand; with the storage bridge inside the confidential VM, the proof is published to S5 and anyone can fetch it and match it to the hash committed on-chain
-  **The whole product on one attested machine (30 September 2026):** chat, FLUX.2 images, a fine-tune with its adapter served back, and six video-generation modes from the Blender extension, every service deployed in one confidential VM; the configuration hash computed in advance and signed into the policy matched on the first boot; a mid-day configuration change was released only under a newly signed policy and a resealed container; three boots, three attested key releases; every job settled on-chain
-  Automatic routing of paid sessions to **tee-attested** hosts
-  Video-generation (LTX 2.3) weights under the same attested release (today they are public weights verified against pinned hashes as they download into the confidential VM, not sealed or broker-released)
-  Approved host measurements separated from the sealed model binding, so a new host type needs no reseal; an independent second signer for those approvals once the node image reproduces from public source

## Private Fine-Tuning (Training M0 — core demonstrated end to end; paid gates pending)

-  Wire contract with the SDK frozen; node accept, staging, content scan, token re-count, slice loop, per-slice proof and settlement, encrypted adapter delivery
-  Serve-back: a finished adapter attaches to an ordinary inference session, isolated to that session and removed at its end
-  Cross-implementation token counting pinned by a frozen fixture and verified on both sides, so a declared count and the node's recount cannot silently diverge
-  **End-to-end fine-tunes on real GPU hardware**, billed exactly per token, with the client's count, an independent count and the node's recount agreeing to the token on every run
-  **Demonstrable behaviour change:** an adapter trained on an invented-fact corpus answers questions the base model provably cannot (the control names the subject as likely fictional), verified against a base-model session on the same host
-  **Cross-host serving:** the adapter trained on one host staged and served on a different operator's GPU, a different hardware generation, isolated per session

- ⌚ Paid settlement gates on testnet: one completed run, one mid-run failure, one cancellation, each billed to the unit
- ⌚ Model registration and allow-list bundle publication
- ✅ **Fine-tuning on the attested confidential-VM host (30 September 2026):** the dataset decrypted only inside the confidential VM, the adapter returned encrypted and served back into a session on the same machine; two independent runs of the same job produced byte-identical adapters
- ⌚ **Confidential training as a guarantee** — the dataset key still travels in the job payload to whichever host the client chose, so fine-tuning is not yet host-blind by construction. Closing it reuses the existing attestation machinery, releasing the dataset key only into an attested enclave, and inherits the GPU-memory question of Section 8.6

## Weight Binding for Third-Party LTX Hosts (Planned — mainnet cut)

*Prerequisite for opening AI video generation to hosts other than the reference operator (Section 4.7, "Precision").*

- ⌚ Per-template **weight manifest** — every checkpoint, LoRA, text encoder, VAE and depth model a pinned graph loads, hashed from disk by the node and reduced to one **weightsRoot**
- ⌚ **weightsRoot** registered on-chain per model id; the client checks it **before escrow**, so a host advertising unregistered weights is refused before any money moves
- ⌚ **Engine inventory** — a content hash of the host's ComfyUI custom nodes, because a pinned graph pins class names, not the code behind them: a custom node that shadows a loader class defeats weight hashing on its own
- ⌚ Host-side binding raises the cost of misrepresentation and leaves an on-chain evidence trail; it is not host-proof. Only the confidential-computing path (Section 8.6) makes it non-forgeable

## Phase 7: Economic Refinement (Planned)

- ⌚ Real-time uptime and latency metrics collection
- ⌚ On-chain reputation scoring
- ⌚ Host quality dashboards
- ⌚ Dynamic pricing mechanisms

## Phase 8: Decentralisation (Planned)

- ⌚ DAO governance launch
- ⌚ Multi-sig transition
- ⌚ Community-driven upgrades
- ⌚ Dispute resolution DAO

## Phase 9: Scaling (Future)

- ⌚ Additional L2 chains
- ⌚ Cross-chain sessions
- ⌚ Optimistic proof batching

- 🕒 State channels for high-frequency
  - 🕒 Multi-orchestrator coordination via A2A
  - 🕒 Agent marketplace with on-chain reputation scoring
  - 🕒 Confidential computing (TEE): production hardening beyond the September 2026 hardware proof, namely attested-host routing in the SDK, and multi-vendor TEE support
- 

## 12. Conclusion

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Platformless AI represents a fundamental shift in how AI services are delivered. By eliminating platform intermediaries and replacing trust with cryptography, the protocol creates an open, permissionless marketplace where:

- **Users** maintain privacy, data sovereignty, and choice
- **Hosts** operate independent businesses with fair compensation
- **SaaS companies** embed AI into products without vendor lock-in or privacy risk
- **Developers** build with autonomous AI agents on private, sovereign infrastructure
- **Innovation** flourishes without gatekeepers
- **Trust** is established through verifiable open-source code, cryptographic mathematics, and carefully designed economic incentives—not platform reputation

The three pillars of trustless operation:

1. **Code:** Open-source smart contracts that anyone can audit, with deterministic execution
2. **Mathematics:** Cryptographic proofs, signatures, and encryption that provide guarantees no human can override
3. **Economics:** Staking requirements, payment escrow, and fee structures that align incentives towards honest behaviour

With the Claude Bridge and OpenAI Bridge extending the protocol to support autonomous AI agents, SaaS AI backends, and image generation, the transcoding pipeline adding encrypted GPU-accelerated video processing with load balancing, and the orchestrator package adding experimental multi-agent coordination via the A2A protocol (v1.0.0-rc) with x402 micropayments, Platformless AI is no longer limited to chat-based inference. The same encrypted, decentralised infrastructure that protects a private conversation also protects an enterprise development team's source code as Claude Code or OpenCode edits it, transcodes video across multiple GPU hosts with HLS adaptive bitrate streaming and per-segment encryption without any intermediary seeing the content, orchestrates multi-agent workflows that decompose complex goals across optimal models, generates images from natural language prompts via FLUX.2 diffusion without exposing creative intent to intermediaries, generates video from a text prompt or the user's own encrypted stills via LTX 2.3 — and reframes, edits, restyles, and restores footage the artist already has, from inside Blender's own timeline, with provenance the buyer's own client verifies (which model, which template, byte-exactly which input images and control clip) and an attestation anchored on-chain at a settlement that pays the host and refunds the remainder automatically — or processes a SaaS product's customer data as it flows through AI-powered features. Since NVENC encoding uses dedicated hardware separate from the CUDA cores, hosts can serve AI inference and video transcoding simultaneously — dual revenue streams from the same GPU without performance degradation. With both Anthropic and OpenAI API compatibility plus early A2A protocol adoption

— the first DePIN project to implement it — virtually any AI-powered application or agent network can migrate to decentralised infrastructure. And for the most sensitive workloads, the confidential-computing path charts the way beyond encryption-in-transit to host-blind inference inside Trusted Execution Environments, now proven end to end on real Intel TDX and NVIDIA H200 hardware, with paid testnet sessions already run on it and one open GPU-evidence question still to answer, so that one day even the host serving a model cannot read the prompt it answers or the weights it runs. Companies are already approaching us to use this infrastructure.

The AI revolution should not be controlled by a few large platforms. Platformless AI provides the infrastructure for a truly open AI ecosystem where anyone — users, developers, enterprises, and SaaS companies — can participate, compete, and benefit.

**Join the revolution. Become platformless.**

## Appendix A: Contract Addresses (Base Sepolia)

| Contract       | Proxy Address                              |
|----------------|--------------------------------------------|
| JobMarketplace | 0xD067719Ee4c514B5735d1aC0FfB46FECf2A9adA4 |
| NodeRegistry   | 0x8BC0Af4aAa2dfb99699B1A24bA85E507de10Fd22 |
| ModelRegistry  | 0x1a9d91521c85bD252Ac848806Ff5096bBb9ACDb2 |
| ProofSystem    | 0xE8DCa89e1588bbdbc4F7D5F78263632B35401B31 |
| HostEarnings   | 0xE4F33e9e132E60fc3477509f99b9E1340b91Aee0 |
| FAB Token      | 0xC78949004B4EB6dEf2D66e49Cd81231472612D62 |
| USDC Token     | 0x036CbD53842c5426634e7929541eC2318f3dCF7e |

## Appendix B: SDK Installation

```
npm install @fabstir/sdk-core
```

```
import { FabstirSDKCore, ChainId } from "@fabstir/sdk-core";

const sdk = new FabstirSDKCore({
  chainId: ChainId.BASE_SEPOLIA,
  rpcUrl: "https://sepolia.base.org",
  contractAddresses: {
    jobMarketplace: "0xD067719Ee4c514B5735d1aC0FfB46FECf2A9adA4",
    nodeRegistry: "0x8BC0Af4aAa2dfb99699B1A24bA85E507de10Fd22",
    // ... other addresses
  },
});

await sdk.authenticate("metamask");
const sessionManager = sdk.getSessionManager();
```

## Appendix C: Resources

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- **Documentation:** <https://docs.fabstir.com>
  - **GitHub:** <https://github.com/Fabstir>
  - **SDK Reference:** [https://github.com/Fabstir/fabstir-llm-sdk/blob/main/docs/SDK\\_API.md](https://github.com/Fabstir/fabstir-llm-sdk/blob/main/docs/SDK_API.md)
  - **Contract Reference:** <https://github.com/Fabstir/fabstir-compute-contracts>
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**Platformless AI** — Trustless AI for Everyone

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